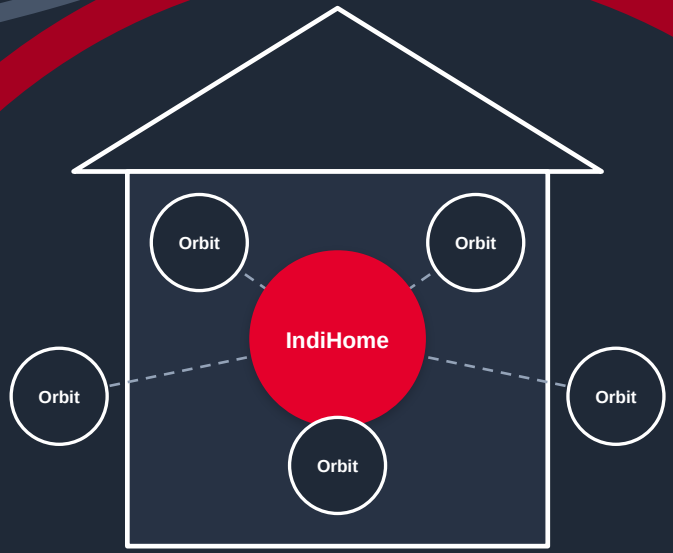
Telkomsel · IndiHome · Orbit

H . **O** . **M** . **E**

Household fit · Offer a path up · Make staying better · Evaluate household value

Managing the home, not the line: positioning **IndiHome** and **Orbit** by **household need** to grow **long-term household value**



IndiHome as the family's home hub; Orbit as the flexible start, in and beyond the home

Proposed by:

Team [name]

[photo] [member 1]

[photo] [member 2]

[photo] [member 3]

Connecting Every Home: Optimizing Telkomsel's Household Connectivity Portfolio · DWDG ITS Mini Case Competition 2026

Executive summary: fitting each home to the right product, then growing and keeping it, is worth **Rp394bn** over the life of the homes won

Situation

Household revenue fell **1.6%** in 2025 and **3.8%** in 1H26 [1] while **≈58–60M** homes are still unconnected [2] and **Rp100k fiber** took the growth: **Surge added 1.98M homes** in 12 months [3].

Complication

H · Household fit	O · Offer a path up	M · Make staying better
Products placed by network and price tier: none of **5** product pages routes a household across products [4]; **0.8–1.2M** inactive accounts came in on loose terms [1]	**Extras are lent:** Double Speed lasts **180 or 360 days** [4]; the average home pays **2.5% below to 8.4% above** the entry price [5]	In cities IndiHome wins **1 of 50** award slots and is **last of 8** on upload [6]

Solution

Household fit	Offer a path up	Make staying better
One **five-question check** in app, web, GraPARI, 188 and agents; **Orbit = flexible start**, **IndiHome = family hub**; opt-in **home check** for Telkomsel families a rival is about to connect; sales paid on homes **still paying at 3 and 12 months**	**IndiHome ladder steps** (single → family, EZnet → IndiHome); a **chosen option before each boost ends**; the **household plan** rewards each added line with quota and services, not a discount	**Home Wi-Fi promise** in contested cities; **retention triggers** with a random hold-out; cheaper plans offered **only to save a home**

Impact (est.)

213k paying homes by 2031 est.	**Rp534bn** over the life of the homes; **Rp415bn** revenue a year by 2031 est.	**204k** homes kept by 2031; **Rp174bn** over the life of the homes est.

E **E · Evaluate household value:** one **household identity** (opt-in), **household KPIs** in the 2027 KPI contracts, **one owner** of household value. Telkom already states 'Maximize LTV' for IndiHome [1]; E extends it across **Orbit and mobile**.

Rp394bn	35.1%	2.95	2031	−Rp46bn	Rp295bn	Rp98bn
NPV over the life of the homes won est.	IRR est.	B/C est.	Cash payback est.	5-year NPV est.	Capex est.	Failed-pilot cash est.

Source: [1] Telkom 1H26 Info Memo · [2] Appendix 6 · [3] Surge disclosures, Appendix 7 · [4] Telkomsel website (10 Oct 2026), Appendix 9 · [5] Appendix 10 · [6] Opensignal, Appendix 8 · model: Appendix 21–24

Background

Background: Telkomsel leads home broadband, yet household revenue is falling while **Rp100k fiber** takes most of the market's growth

1 **Market:** where household connectivity is heading

New fixed homes in 12 months (millions)

Whole market, a year (forecast) [5]



+1.28M

Surge alone, 2Q25 → 2Q26 [6]



+1.98M

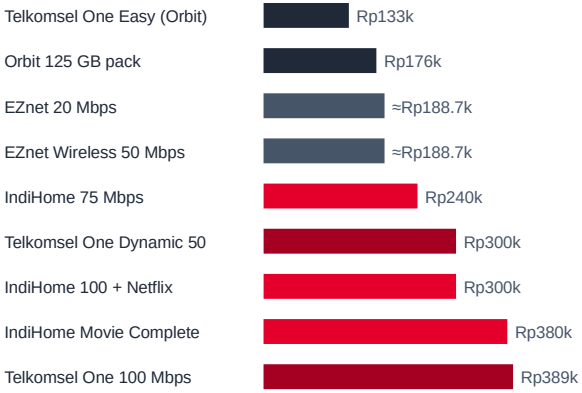
≈58–60M homes still have **no fixed broadband**: penetration is only **18–21%** of 73M households [5]

The Rp100k anchor [4]

Starlite fiber (since 2025) and licensed **1.4 GHz** fixed wireless (since Feb 2026) sell **100–200 Mbps for Rp100k**, VAT included, with a free modem. IndiHome's entry plan costs **2.4×** that.

2 **Telkomsel's household portfolio:** five propositions

Nine offers between ≈Rp130k and ≈Rp390k a month [7]



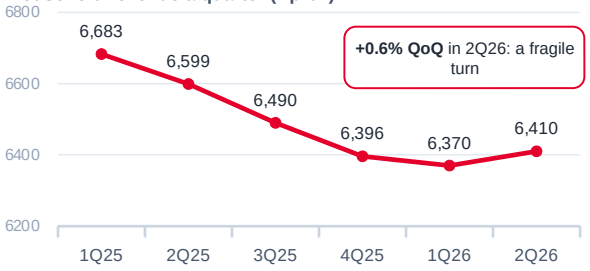
VAT included, Jakarta Selatan, Oct 2026. Priced per GB, per Mbps or per bundle, on prepaid, monthly or 12-month terms.

153.5M mobile customers [1]

≈67% of fixed broadband connections [3]

3 **Performance:** customers and revenue

Household revenue a quarter (Rp bn) [1]



- Base **9.513M** homes in 2Q26, **−5.4%** YoY [1]
- 2025: **+0.7M** customers reported, yet **−Rp0.44T** of revenue [2]
- **0.8–1.2M** 'inactive, non-revenue-contributing' accounts removed or identified [1]

The case's Key Question [8]

*"As household connectivity needs become increasingly diverse, how should Telkomsel **differentiate and position IndiHome and Orbit across customer segments** to unlock stronger **long-term household value** without relying primarily on price competition?"*

Source: [1] Telkom 1H26 Info Memo · [2] Telkomsel AR 2025 · [3] Opensignal, Nov 2025 · [4] Starlite and Internet Rakyat pricing (Bisnis, Kontan) · [5] Media Partners Asia; Komdigi · [6] Surge (WIFI) disclosures · [7] Telkomsel app and website, 10 Oct 2026 · [8] Case book · Appendix 2, 6

Problem Identification

Problem identification: IndiHome and Orbit are placed by network and price tier, and household value leaks at **entry**, in **spend** and in **delivery**

The problem, stated twice

Symptoms [1]

In 2025 Telkomsel reported **0.7M more** household customers but lost **Rp0.44T** of household revenue; in 2026 it began removing **0.8–1.2M** inactive accounts, and revenue was still **3.8% lower** in 1H26.

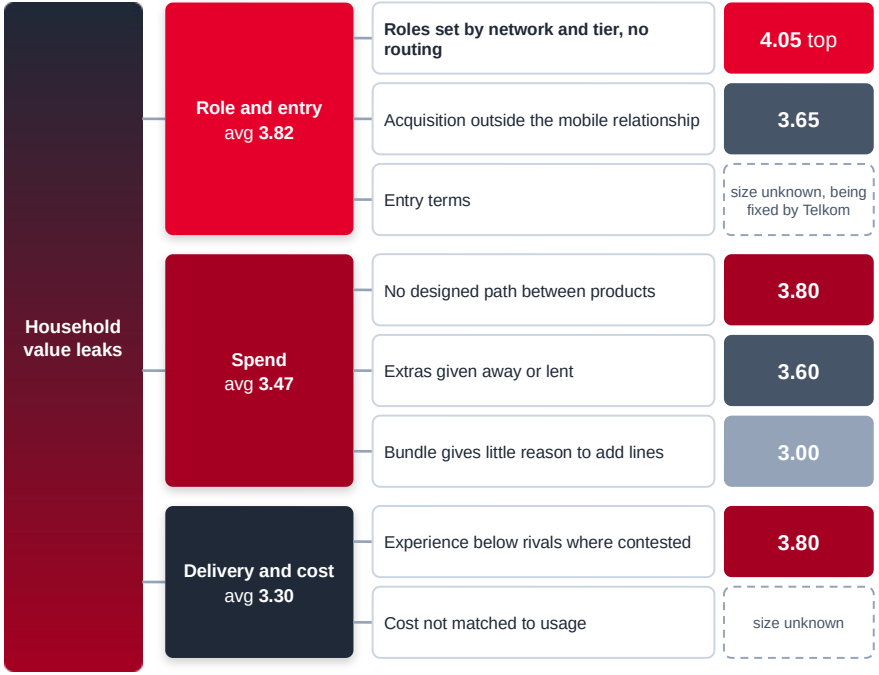
Key Question

Why is Telkomsel losing value per household in a growing market, and how much of it comes from the way **IndiHome and Orbit are positioned**?

Causes, after the analysis [3]

Products are assigned by **network and price tier**, the extras that should raise spend are **lent**, and the premium is **unproven where rivals compete**. Underneath, LTV is Telkom's stated aim, yet **no one is measured on what a household is worth** across IndiHome, Orbit and mobile.

Issue tree: 3 branches, 8 drivers, each scored [3][4]



Role and entry ranks first in every test with the unscored drivers at their midpoints (**17 of 20** tests at their extremes).
Score = significance 40% · urgency 25% · controllability 20% · evidence 15% [4]

What holds in every reading

−0.7% to −10.8%

change in revenue per **paying** household in every timing scenario for the inactive accounts est. [2]

−899k vs +85k

2Q26 net change: homes **without** a Telkomsel line vs **converged** homes (net figures; the removed accounts are not published) [1]

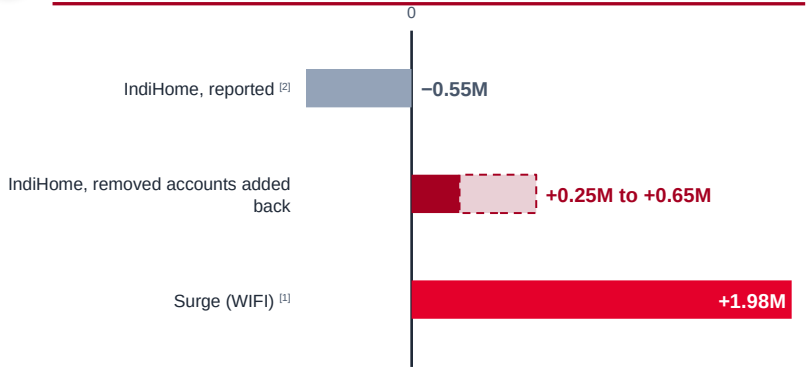
So the solution order follows the ranking: fix the roles of IndiHome and Orbit first, then keep homes and build paths up, all measured on household value.

Source: [1] Telkom 1H26 Info Memo · [2] Appendix 3, 4 (cleansing, scenarios) · [3] Appendix 14 (issue tree) · [4] Appendix 16 (scoring and sensitivity)

Analysis: Market and Competition

Market and competition: Rp100k fiber took the growth, with **Surge adding 1.98M homes** in a year while IndiHome's reported base fell by **0.55M**

1 **Same-window growth**, 2Q25 → 2Q26 (homes)



78% of Surge's base is fiber; its growth was ≈3–8× IndiHome's underlying growth (Appendix 7)

So what: a two-front squeeze [1][3]

Price front (≈Rp100k): Starlite fiber, Internet Rakyat, MyRepublic Air, HiFi Air. Surge **cut its 2026 wireless target from 3M to 1M** and moved up-market (**500 Mbps for Rp168k, 2 Gbps for Rp250k**), so the low-price threat is fiber more than wireless.

Quality front (Rp345–482k): XL SATU, Indosat HiFi, Biznet, Oxygen.id win measured experience in their strongholds. IndiHome sits in between, with **reach** as its distinct ground.

2 **Measured experience in cities** (Opensignal, urban view) [3]

Metric	IndiHome	Rank of 8 ISPs (1 = best)	vs leader
Download speed	**26.4 Mbps**	7	**−35.3%**
Upload speed	**13.3 Mbps**	8	**−56.4%**
Consistent quality	**58.8%**	4	**−15.9%**

1 of 50 urban award slots outside Maluku–Papua

18 of 45 rural award slots outside Maluku–Papua

30 of 30 award slots in Maluku–Papua: the only scale ISP

3 **Could something else explain it?** Four of eleven rival explanations

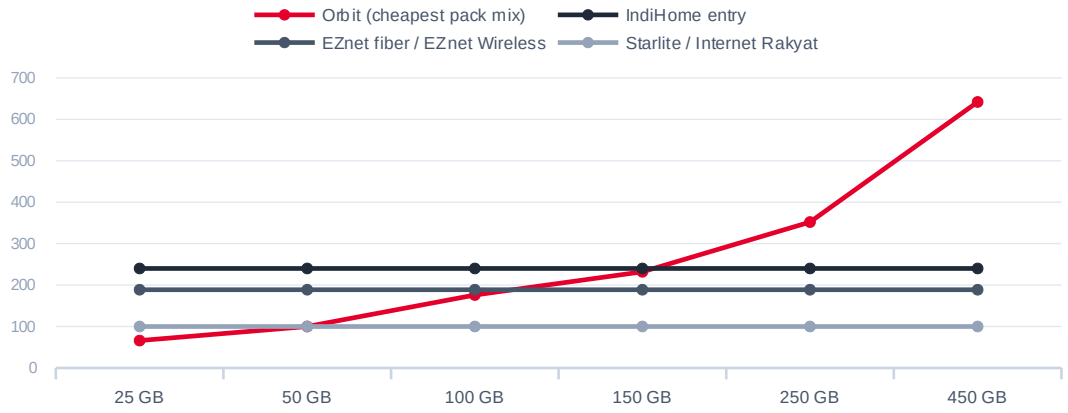
Macro squeeze	Does not explain 2026: mobile revenue **+5.6%** while household revenue fell [2]
Bundle revenue allocation	Cannot be tested with public data
Convergence follows engagement	Cannot be ruled out, so the acquisition-channel driver **stays a hypothesis**
Shift to single-play	Partly true, a mechanism **inside spend**

All 11 explanations are tested in Appendix 5

Source: [1] Surge (WIFI) disclosures; Samuel and BCA Sekuritas · [2] Telkom 1H26 Info Memo · [3] Opensignal, Jan–Mar 2026 · Appendix 5, 7, 8

Households and portfolio: Orbit is Telkomsel's cheapest home option only up to about **125 GB** a month, yet no product page routes a family to it

1 What each option costs a household a month, by monthly use (Rp k) [1][2]



1. Orbit costs more than **EZnet** above **95–125 GB** a month
2. Orbit costs more than **IndiHome** above **~155 GB**
3. At 450 GB Orbit costs **Rp642k, 2.7× IndiHome**

*Phone data alone costs **Rp78k at 25 GB** and **Rp465k at 150 GB** at Telkomsel's average yield, so a home line pays from ~25–35 GB [3]. EZnet Wireless slows to 5 Mbps above 350 GB.*

2 The family that matters most

≈110–255 GB — a month for a streaming family of 4–6 on a conservative build-up est. [3][4]: **above Orbit's cheap zone**, yet Orbit's page says it is "for everyone" [1]

Home Wi-Fi moves phone data off the network whoever connects the home

- **≈Rp117–156k** — a month of phone data in a 3–4-line Telkomsel family est.
- **50–70%** — of it moves to home Wi-Fi once the home is connected est.
- **≈Rp58–109k** — a month of mobile revenue lost when a rival connects the family est.

The routing check: five Telkomsel product pages, read 10 Oct 2026 [1]

IndiHome	Orbit	EZnet	EZnet Wireless	Telkomsel One
✖ no cross-product tool	✖ no cross-product tool	✖ no cross-product tool	✖ no cross-product tool	✖ no cross-product tool
"Cek Jangkauan" shows IndiHome plans only	*"Paket modem WiFi spesial untuk semua"*	*"Internet rumah murah dan mudah untuk semua"*	*No comparison with Orbit, its twin job*	*"Untuk keluarga", with no plan-to-family match*

Source: [1] Telkomsel website and app (10 Oct 2026) · [2] Rival price pages: Starlite, Internet Rakyat (Bisnis, Kontan) · [3] Telkom 1H26 Info Memo (data yield) · [4] Netflix Help Center · Appendix 9, 11, 13

Analysis Conclusion

Analysis conclusion: product roles are the **biggest lever**, with keeping homes and building upgrade paths close behind

Current Condition	Key Question	Problem	Ideal Condition
Products placed by network and price tier; none of 5 pages routes a home; 0.8–1.2M inactive accounts, mostly outside the mobile base	Why is Telkomsel losing **value per household** in a growing market, and how much of it comes from the way **IndiHome and Orbit are positioned**?	**Role and entry:** wrong product, wrong channel, loose terms	Every home routed by need to the product that fits, through the mobile relationship, on paying terms
Average home pays near the entry price; speed lent for 180–360 days; the bundle is at best a small saving		**Spend:** extras lent, no designed path up	Household spend rises because upgrading and adding lines pay off for the family
1 of 50 urban award slots; upload last of 8; fixed complaints resolved 94.23% vs 98.37% for mobile		**Delivery and cost:** premium unproven where rivals compete	Measured experience at or above rivals where they compete; cost matched to usage
ARPU counts inactive accounts; convergence counted as overlap; Orbit reported inside mobile		**Measurement:** no one owns household value	Household lifetime value is the operating metric, with one owner

Which problems to tackle?	Which solutions?	Why?
Role and entry	H Household fit	Top driver; most urgent; fully in Telkomsel's control
Spend	O Offer a path up	Upgrade steps carry most of the value
Delivery	M Make staying better	Homes kept are the second-largest value
Measurement	E Evaluate household value	Makes the other three judged on household value

Source: Appendix 14 (issue tree, five whys), 15 (gap analysis), 16 (prioritisation)

Strategy overview: H.O.M.E. fits each home to the right product, offers a path up, makes staying better and measures household value

H O M E H.O.M.E.: manage the home, not the line

	H Household fit	O Offer a path up	M Make staying better
What	Put every home on the product that fits its need, through the family's Telkomsel relationship, on paying terms	Give each home a path up that it chooses, so spend rises with the household	Keep homes where rivals are strongest by fixing the home experience and saving homes before they leave
How	**Five-question check** in every channel; **stated roles** (Orbit = flexible start, IndiHome = family hub); **opt-in home check**; pay sales on homes **paying at 3 and 12 months**	**Usage-triggered ladder offers**; **boost-end choice** with 30 days' notice; **household plan** that rewards each added line	**Wi-Fi check**, mesh for **75k at-risk homes** a year, **fix-time promise**; **triggers with a hold-out**; route down **only to save**
Why	Top driver; a rival-threatened family pays from **month one**	Single → family **Rp213bn**, EZnet → IndiHome **Rp121bn** of life value	Keeping a home is cheaper than winning one
Impact	**213k** paying homes by 2031 est.	**Rp534bn** life value est.	**204k** homes kept by 2031 est.

E **E · Evaluate household value (the foundation):** one household identity (opt-in) · household KPIs in the 2027 KPI contracts · one owner of household value

Lasting value, responsibly

- One VAT-inclusive price everywhere
- No silent end to a boost
- First connection without commitment
- Parental controls in the household plan
- Orbit modems refurbished

Source: Appendix 18 (roles), 19 (initiatives), 20 (gates and KPIs), 24 (value split), 26 (theme)

Solution · H: Household fit

Product roles and routing: Orbit becomes the **flexible start**, IndiHome the **family's home hub**, and one five-question check routes every household

1 **Priority households** and the product each needs

Household need	Ambition	Lead product
Families of 4–6+, streaming, children (46.5% of homes have 4–5 people [3])	**Win and grow**	IndiHome hub + household plan
Performance: gaming, hybrid work	**Grow**	IndiHome higher tiers + Wi-Fi promise
Large or premium homes	**Grow**	IndiHome + mesh or FTTR
Budget first connection	**Serve selectively**	Orbit start; EZnet only where a spare port pays
Renting or moving (21% of Jakarta homes rent [3])	**Serve and keep**	Orbit; EZnet Wireless for heavy use
Fiber-dark homes	**Serve**	EZnet Wireless or Orbit, by capacity
Home micro-business	**Route out**	IndiBiz (756k subscribers [1])

What Telkomsel stops doing
- Selling IndiHome, Orbit and EZnet Wireless as interchangeable answers
- Paying acquisition on activation alone
- Letting a boost end without a choice
- Pricing on four VAT bases
- Offering a cheaper plan to sell, instead of to save a home that is leaving

2 The five-question household check

1 Stay at this address 12+ months? ▶ **2** Fiber with a spare port? ▶ **3** Usage up to ~125 GB a month?

	Up to ~125 GB	Above ~125 GB, or streaming, work or gaming
Not staying	**Orbit**	**EZnet Wireless**
Staying, no fiber	**Orbit**	**EZnet Wireless**
Staying, fiber	**EZnet**	**IndiHome**

4 6+ devices or a large home → **Wi-Fi check, mesh or FTTR**

5 2+ Telkomsel lines → **household plan**

Runs in the app, web, GraPARI, 188 and agents; offers rest on consent (PDP law); one VAT-inclusive price.

3 **One household's path**, contribution a month est.

Step	vs staying mobile-only	vs a rival connecting
Orbit start	**−Rp24.9k**	**+Rp19.4k**
On IndiHome	**+Rp2.2k**	**+Rp46.5k**
Plus a household-plan line	**+Rp19.8k**	**+Rp19.8k**

So the home check targets families a rival is about to connect. A Telkomsel family of three lines (Rp138k of mobile revenue) pays from **month one** when a rival would otherwise take the home; the value then comes from the steps up to IndiHome and the added lines.

How Telkomsel wins without leading on price

1. **Fit** — One check across five products; Orbit's no-commitment start
2. **Family relationship** — **153.5M** mobile customers [1]; the household plan
3. **Home experience** — Whole-home Wi-Fi; fix-time promise
4. **Reach** — Widest footprint; the only scale ISP in Maluku–Papua [2]

Source: [1] Telkom 1H26 Info Memo · [2] Opensignal · [3] BPS via GoodStats · Appendix 11 (usage crossover), 13 (household path), 18 (roles and tree)

Implementation: **eight weeks** to validate, **six months** of randomised tests, and national scale only after each gate's pass mark is met

1 Who owns what

Household P&L owner (CEO office)
household KPIs, one budget, shared bonus pool

VP Home Broadband & FMC Consumer Marketing
fit check, roles, boost-end choice, household plan

VP Household Consumer Business Strategy Mgmt
home check, ladder offers, triggers (with CVM)

Billing and IT
boost-end test

Network and CPE
Wi-Fi promise, mesh

InfraNexia
port data, port fill

Finance
household KPIs, cost cap

Critical path: Phase 0 decisions (KPI contracts, P&L owner, InfraNexia data) → **one household identity** → home check, household plan and triggers. In parallel: the fit check and the **boost-end test**, which needs only billing changes and starts first.

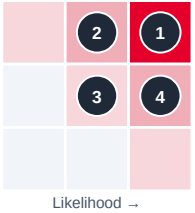
2 **Roadmap** with gates in the model's own units

	Phase 0 · Nov–Dec 2026	Phase 1 · Jan–Jun 2027	Phase 2 · Jul 2027–Dec 2028	Phase 3 · 2029–2031
H	Mystery shop; stated roles; one price	Fit and home check pilot: Jakarta Selatan + a fiber-dark area	National fit and home check	Thresholds tuned on real use
O	Size the expiring boost cohorts	National randomised boost-end test; ladder and plan offers vs hold-out	Paths and household plan nationwide	Household value in targets
M	Household survey	Wi-Fi promise pilot; triggers vs hold-out	Wi-Fi promise in contested cities from 2028	Port-fill terms with the group
E	KPI contracts; P&L owner; InfraNexia data; consent	Household identity build	Household KPIs monthly; Orbit baseline	Household value in pay

Gate 1 Jun 2027 · **Gate 1b** Dec 2027 / Jun 2028 · **Gate 2** Dec 2028

Gate 1 pass mark: every take-up KPI ≥ 60% of plan, e.g. boost-end take-up ≥ **4.8%** · save rate ≥ **6%** · opt-in × conversion ≥ **0.6%** · ladder steps ≥ **1.8% / 0.3%** · lines ≥ **1.2%**. If it fails, national spend stops.

3 Top risks and triggers



#	Risk	Trigger → action
1	Organisation resists household KPIs	Not in 2027 KPI contracts → Phase 0 does not exit; escalate to the CEO office
2	Cannibalisation (routing down)	Leakage > 5% of new homes → tighten price fences
3	Boost-end backlash	Take-up < 4.8% at month 3 → switch to outcome packs; never a silent cut
4	Rival price moves	Rival build detected → triggers and Wi-Fi checks there first; no price cut

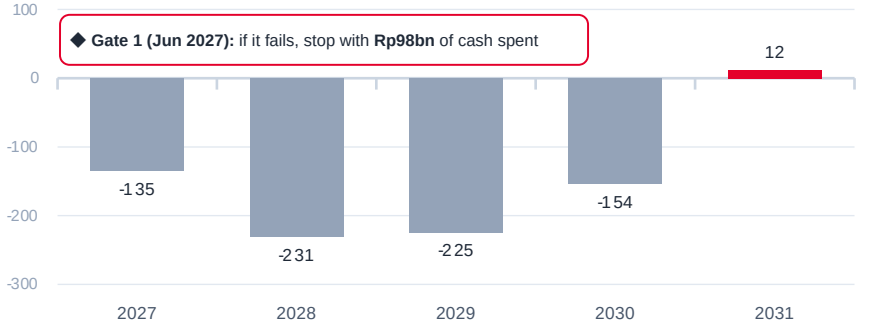
Source: Appendix 19 (initiatives and RACI), 20 (gates and KPIs), 25 (risks and feasibility). VP titles as reported in Bisnis; other functions generic.

Financial Projection

Financial projection: cash pays back in **2031** and the homes won are worth **Rp394bn** after all capex, while a failed pilot stops at about **Rp98bn**

1 **Cash and value**, Rp bn est.

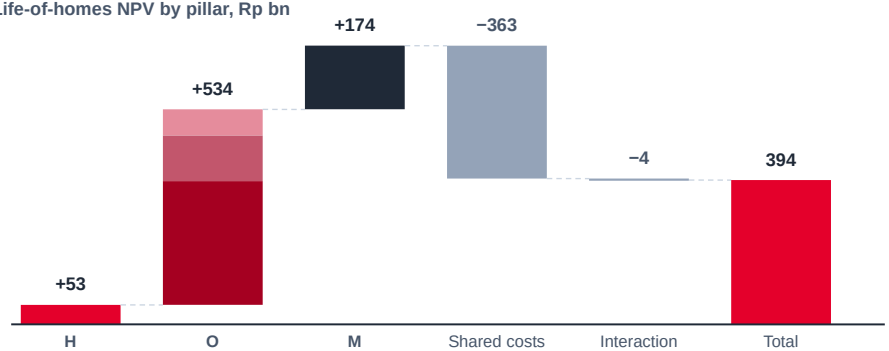
Cumulative free cash flow against the do-nothing base



NPV, life of homes won	Rp394bn	IRR	35.1%	B/C	2.95
BEP (cash)	2031	5-year NPV	−Rp46bn	Capex 2027–31	Rp295bn

2 **Where the value comes from** and how robust it is est.

Life-of-homes NPV by pillar, Rp bn



*O = ■ **paths Rp338bn** (ladder steps 334 + Orbit → fiber 4) · ■ **lines Rp125bn** · ■ **boost-end Rp71bn**. Shared costs = platforms, marketing, programme office.*

Robust: value falls to zero only if every take-up input lands at **55% of plan** at once; Gate 1 passes at 60%.

Gated: expected NPV **Rp208bn** at a 60% chance of passing Gate 1.

Price-led comparator: every **Rp10k** cut across the base costs **≈Rp1,142bn a year**, nearly all of it margin, over five times the **Rp210bn** of EBITDA a year this plan adds by 2031.

Measured by (net stocks est.), 2028 → 2031

73k → 213k	59k → 204k	67k → 197k	90k → 267k	Rp1,134bn
paying homes won	homes kept	Telkomsel lines added	ladder steps	revenue a year by 2031 (≈4.4%)

Conclusion: H.O.M.E. answers the case without a price war: each home gets the product that fits, a path up it chooses and a reason to stay, all judged on household value and released only through gates.

Source: Model in Appendix 21–24: incremental to a do-nothing base, Rp bn ex-VAT, 10% discount rate, 22% tax, life of homes won = 2027–2041 run-off with no new homes after 2031

Appendix

The evidence behind every core claim, grouped by the criterion it serves. Core footnotes point here.

Case and scoring map

A1 Case requirements and scoring matrix · p.12

Problem identification

A2 Household revenue, base and ARPU · p.14
A3 The 2Q26 cleansing · p.15
A4 Paying households · p.16
A5 Rival explanations · p.17
A14 Root cause · p.27
A15 Gap analysis and governing cause · p.28
A16 Prioritisation · p.29

Analysis and research

A6 Market sizing and value pools · p.19
A7 Competitors · p.20
A8 Measured experience · p.21
A9 Digital mystery shop · p.22
A10 Spend · p.23
A11 Usage crossover · p.24
A12 Households · p.25
A13 Offload · p.26
A17 SWOT with a so-what and the TOWS moves mapped to H, O and M · p.30

Solution

A18 Product roles and the routing tree · p.31
A19 Initiatives and owners · p.32
A26 Theme · p.41
A28 Considered and parked · p.44

Feasibility, model and risk

A20 Gates and KPIs · p.33
A21 Financial model · p.34
A22 Financial results · p.36
A23 Sensitivity · p.37
A24 Value split · p.38
A25 Feasibility and trade-offs · p.39
A27 Assumptions register · p.42

References

R1 References · p.45
R2 References · p.46
R3 References · p.47

Appendix 1 (1/2)

Case requirements: every clause of the case question mapped to where the deck answers it

✓ **One integrated household strategy** → P7 · A19	✓ **Priority customer segments** → P8 · A12	✓ **Each segment's connectivity needs** → P5, P8 · A11, A12	✓ **Role of IndiHome and Orbit per segment** → P8 · A18
✓ **Products not treated as interchangeable** → P5, P8 · A11	✓ **Balance of acquisition, affordability, retention, monetisation, experience, differentiation** → P7 · A25 (2/2)	✓ **Levers: positioning, migration paths, cross-sell, bundling, FMC, digital services, partnerships** → P7, P8 · A19	✓ **Commercially feasible in Indonesia** → P10 · A22, A24
✓ **Operationally feasible** → P9 · A25	✓ **Implementation roadmap** → P9 · A20	✓ **Risks and mitigation** → P9 · A25	✓ **Measurable KPIs** → P9, P10 · A20
✓ **Business impact: conversion, ARPU uplift, retention, FMC adoption, incremental revenue** → P10 · A22	✓ **Without relying primarily on price competition** → P8, P10 · A24	✓ **Stronger long-term household value** → P10 · A21, A24	✓ **Increasingly diverse household needs** → P5 · A11, A12

The four Key Questions for participants

KQ1 Which segments, with what needs → P8 (priority households) · A12	**KQ2 The role of IndiHome and Orbit** → P8 (check and roles) · A18	**KQ3 Acquire, engage, retain, grow** → P7 (H, O, M) · A19	**KQ4 Implement and measure** → P9, P10 · A20–A24

Source: Case book (S01, p.3–4); guidebook (S02)

Appendix 1 (2/2)

Scoring matrix: where each criterion is answered, weighted as the judges weight it

Executive Summary	**5%**	Concise summary; problem, solution and outcomes communicated	**P1 (SCSI matrix, number row)**
Problem Identification	**15%**	Clear problem; depth of understanding; issues prioritised by significance and impact	**P3 (issue tree, scores), P6; A3–A5, A14–A16**
Analysis and Research	**25%**	Relevant research; quality of data, tools and methods; opportunities, risks, threats and strengths	**P2, P4, P5; A2–A13, A17**
Solution Quality	**30%**	Critical, strategic reasoning; creative solutions consistent with the analysis	**P6 bridge, P7, P8; A18, A19, A26**
Feasibility and Risk	**20%**	Realistic, sequenced plan; sound financial model tied to operations; risks, trade-offs, contingencies	**P9, P10; A20–A25, A27**
Formatting	**5%**	Format compliance; clear professional business English	**10 core pages, 16:9 PDF, English, references (R1–R3)**

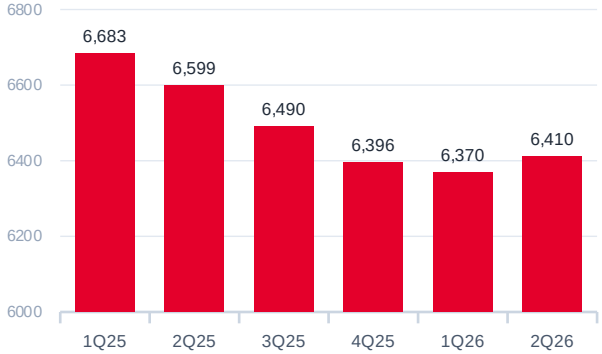
Half the marks (Solution 30% + Feasibility 20%) sit on the answer and its execution, so the solution is built from the analysis (P6) and every model figure ties to a gate (P9) and an appendix page (A21–A24).

Source: Guidebook, scoring matrix (S02, p.10)

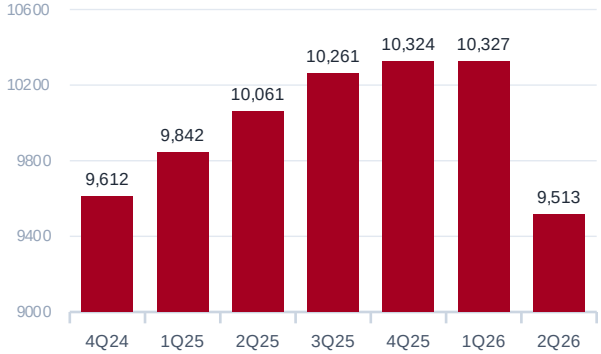
Appendix 2

Household revenue, base and ARPU: revenue fell in 2025 and 1H26 while the reported base shrank 5.4%

Household revenue a quarter (Rp bn)



Reported base, thousands of homes



Revenue bridge 2024 → 2025 (Rp bn)

Step	Rp bn
FY2024 revenue	26,606
+ volume (avg base 9.153M → 9.968M)	**+2,369**
− price and mix (Rp242.2k → Rp218.8k)	**−2,807**
FY2025 revenue	**26,168** (−1.6%)

ARPU on three bases, 2Q26 (Rp k a month)

Base	ARPU	vs cheapest standalone plan ex-VAT (Rp216.2k)	Basis
As reported (implies ≈10.14M accounts)	**210.8**	**−2.5%**	Telkom 1H26 Info Memo
On the 9.513M closing base	**224.6**	**+3.9%**	our calculation
On a paying base under the 0.8M reading (≈9.11M)	**234.5**	**+8.4%**	estimate

Reading: whichever base is used, the average IndiHome home pays between 2.5% below and 8.4% above Jakarta Selatan's cheapest standalone plan, so upsell depth is shallow.

Annual series

ARPU **Rp253k → Rp237.6k → Rp213.5k** (2023–2025), −15.6% in two years.

Revenue **Rp26.6T → Rp26.2T** (2024–2025); 1H26 **Rp12.78T, −3.8%** YoY.

Mobile ARPU fell **3.1%** in 2025 but rose **9.0%** YoY in 2Q26; mobile revenue **+5.6%** in 1H26.

2Q26 +0.6% QoQ is helped by removing inactive accounts and World Cup marketing: one quarter is not a turn.

Source: Telkomsel AR 2025 (S03); Telkom info memos FY24–1H26 (S04–S08); our calculation

Appendix 3

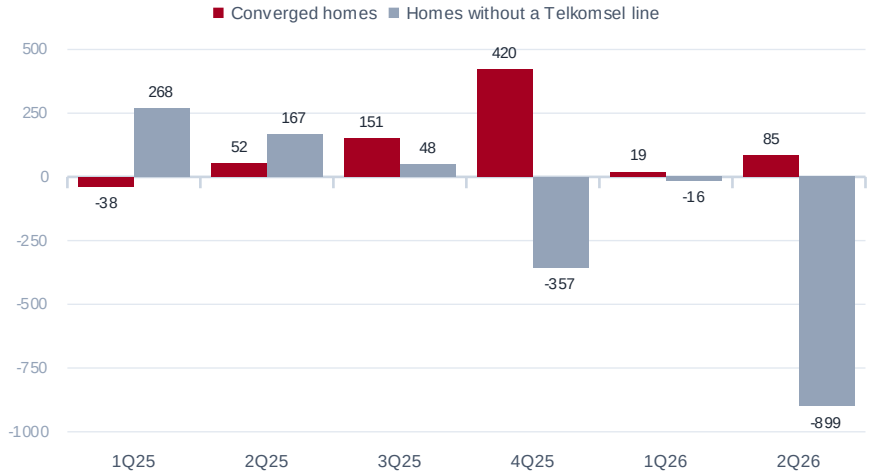
The 2Q26 cleansing: two readings of its size, and where the net fall sat

Two readings of the same event

	Telkom's memo	BRI Danareksa
What it says	"~1.2 Mn · inactive, non-revenue-contributing customers with a LoS of >12 months"; one-off ~Rp130bn non-cash adjustment	"cleansed ~800k of non-contributing subs"; phased cleansing; FY26F base 9.124M
Implied 2Q26 underlying net adds	**+386k** (vs +3k in 1Q26)	**−14k**, in line with 1Q26
Inactive accounts still in 9.513M	~0	**~0.4M**
Share of net adds Jan 2024–Jun 2025 (1.367M)	**87.8%**	**58.5%** removed so far
Share of the 1Q26 base	**11.6%** (1 in 9)	**7.7%** (1 in 13)

*The slide does not say the accounts never paid; it says they were older than 12 months and brought in no revenue. The deck carries both readings as a range: **0.8–1.2M**.*

Quarterly change in homes (thousands): converged vs without a Telkomsel line



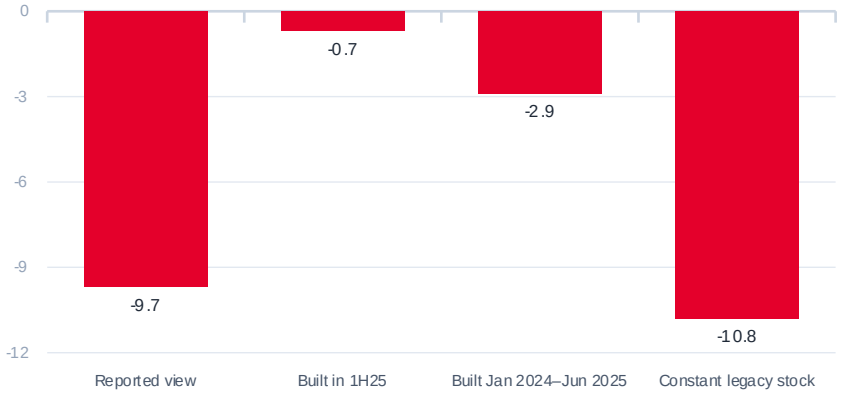
What the series shows
· In 1H25 almost all net growth was in homes **without** a Telkomsel line (+449k base, ≈+13k converged).
· In 2Q26 the net fall sat in that pool: **−899k vs +85k** converged (net; which accounts were removed is not published).
· The 4Q25 jump (+420k converged on +63k net adds) may be better matching after one bill; 3Q25 is derived from a rounded rate.

Source: Telkom 1H26 Info Memo p.43 (S04); BRI Danareksa, 5 Aug 2026 (S09); Telkom info memos (S05–S08); our calculation

Appendix 4

Paying households: revenue per paying home fell in every timing scenario, and the timing test with its break-even

Change in revenue per household, 2024 → 2025, by when the inactive stock built up (%)



Scenario	Paying growth	Per paying home	Volume / price-mix (Rp bn)
Reported view	+8.9%	−9.7%	+2,369 / −2,807
Built in 1H25	−0.9%	−0.7%	−247 / −191
Built Jan 2024 → Jun 2025	+1.3%	−2.9%	+350 / −788
Constant legacy stock	+10.2%	−10.8%	+2,727 / −3,165

Every row adds to the same −Rp438bn. Scenarios use the 1.2M stock; 0.8M narrows the spread.

Timing-based test: stock held constant from mid-2025

Quarter	Revenue (Rp bn)	Per paying home, 1.2M stock	Same, 0.8M stock
3Q25	6,490	Rp241.4k	Rp231.1k
4Q25	6,396	Rp234.5k	
1Q26	6,370	Rp232.7k	Rp222.9k
Change 3Q25 → 1Q26		**−3.6%**	**−3.5%**

0.33M new non-payers between 3Q25 and 1Q26 would erase the fall (0.33–0.34M by stock); provisions were still rising in 1H26, so the test is **evidence that falls short of proof**.

What holds in every reading
· Household revenue fell: **−1.6%** in 2025, **−3.8%** in 1H26.
· Revenue per paying household fell in all three timing scenarios (**−0.7% to −10.8%**).
· The likeliest timing (built Jan 2024 → Jun 2025) means little real volume growth (+1.3%) and a moderate price-mix fall (−2.9%).
· The cleansing is a robustness check; the headline is the role mismatch.

Source: Telkomsel AR 2025 (S03); Telkom info memos (S04–S08); press timing (S10); our calculation and simulation

Appendix 5 (1/2)

Rival explanations: eleven alternatives tested against the evidence (1/2)

Explanation	If it were the cause, we would see…	What the numbers show	Verdict
R1 · Macro squeeze	All household telecom spend falling	2025: mobile ARPU also fell 3.1%; 2026: mobile ARPU +9.0%, mobile revenue +5.6% while household revenue fell 3.8%	Not for 2026
R2 · A price war	IndiHome cutting headline prices	No headline cut (100 Mbps Rp270k, below rival fiber at Rp345–482k); speed and content added at the same price; Starlite at Rp100k since 2025	Trigger only
R3 · Accounting: only the cleansing	Revenue flat, only counts and ARPU moving	Revenue fell −1.6% (2025) and −3.8% (1H26); per paying home −3.5% in two quarters with the stock constant	Rejected as whole story
R4 · Saturation	Few new fixed homes anywhere	Penetration 18–21%; market +1.28M a year; IndiHome + Surge added 2.2–2.6M in 12 months	Rejected
R5 · Shift to fixed wireless	New homes choosing wireless	78.3% of Surge's base is fiber; fiber is >90–94% of connections; Surge cut its wireless target from 3M to 1M	Rejected
R6 · Industry-wide ARPU slide	Rivals' ARPU falling alike	Surge's blended ARPU is Rp70–108k; XLSmart and Indosat fixed ARPU not published	Partial

Source: File 02 §3: Telkomsel AR 2025 (S03); Telkom 1H26 Info Memo (S04); Opensignal (S14); Surge disclosures (S48, S58, S60); MPA (S24)

Appendix 5 (2/2)

Rival explanations: eleven alternatives tested against the evidence (2/2)

Explanation	If it were the cause, we would see…	What the numbers show	Verdict
R7 · IndiHome is a worse network	Losses where the quality gap is largest	1 of 50 urban vs 18 of 45 rural vs 30 of 30 Maluku–Papua award slots; 4th of 8 on urban consistent quality; churn by area not public	Plausible for urban
R8 · Hidden churn	Non-payers spread across tenure and home types	Stock built during a volume push with activation-paid incentives and no deposit; gross adds not disclosed	Partly true
R9 · Bundle revenue allocation	IndiHome ARPU falling fastest where convergence is highest	Pattern fits (mobile up, fixed down, 59–65% converged) but the allocation rule is not disclosed	Not testable
R10 · Convergence follows engagement	Converged counts jump when matching changes	4Q25: +420k converged on +63k net adds; the 2Q26 split cannot tell the stories apart	Cannot rule out
R11 · Shift to single-play	Revenue per home falls with mix	Each 2P → 1P move is −Rp125k a month; ~525k moves a year would explain the likeliest price-mix fall	Partly true, in spend

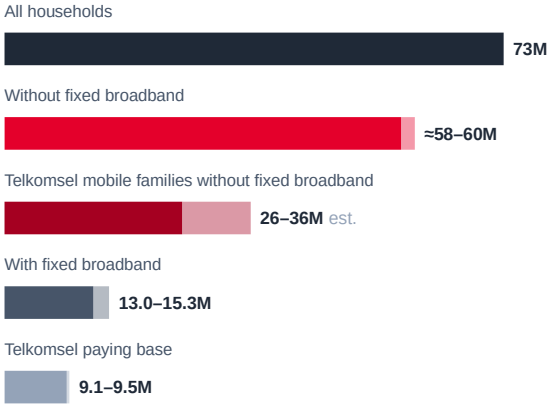
Conclusion: external forces set the Rp100k price point and squeezed 2025 spending, yet none explains the combination of 0.8–1.2M inactive accounts outside the mobile relationship, homes paying near the entry price, a bundle with little saving and no routing across five propositions. Those four are internal and controllable.

Source: File 02 §3: Telkomsel AR 2025 (S03); Telkom 1H26 Info Memo (S04); Opensignal (S14); Surge disclosures (S48, S58, S60); MPA (S24)

Appendix 6

Market sizing and value pools: where the unconnected homes are and what each pool is worth

Households (millions)



Who the next homes are (est.)
Today's fixed market is roughly the middle and upper classes (≈12.9M homes). The next ~20M homes must come mostly from the **aspiring middle class (≈36.0M)**, whose budgets sit near the **Rp100–150k** the regulator uses as the rural household telecom budget. Policy targets **50% by 2029** (≈5.3M homes a year) against a market forecast of **+1.28M a year**.

Value pools Telkomsel could capture

Pool	Size	Value signal	Right to win	Appeal
Telkomsel mobile families without fixed broadband	26–36M est.	≈Rp117–156k a month of phone data, 50–70% of which moves to home Wi-Fi	High	High
Fair share of market growth	794–934k homes a year	≈Rp2.0–2.4T a year at today's revenue per account	Medium	High
Convergence depth in existing homes	3.345M homes without a line	Up to ≈Rp1.85T a year, higher margin	High	High
Reactivation of removed homes	0.8–1.2M, already wired	Not sized	Medium	Medium
Idle fiber	~40% utilised; ~1.2M released ports	Lower cost per extra home if terms allow	High (group)	Medium
Household outcomes (Wi-Fi, safety, gaming)	86% of 1,000 parents interested in controls	≈Rp183–337bn a year per 1% FTTR attach	Medium-high	Medium
Home micro-businesses	IndiBiz 756k	Not sized	Medium	Route out

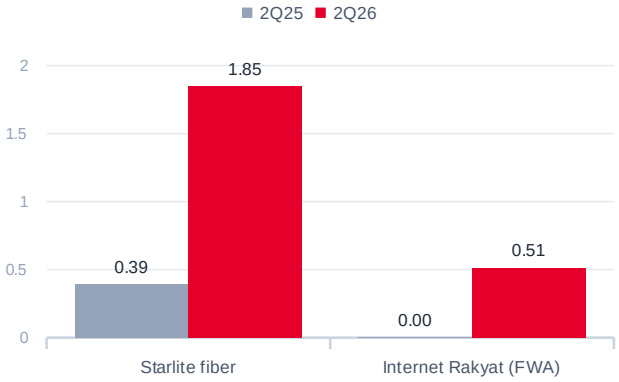
Rp100k anchor: a Rp100k fiber line pays only below ≈Rp90k of incremental cost; budget homes are a cost-to-serve question, served selectively (P8).

Source: Media Partners Asia (S24); Komdigi (S21); BPS via GoodStats and Mandiri Institute (S50, S51); Telkom 1H26 Info Memo (S04); our calculation and estimate

Appendix 7

Competitors: prices, measured quality and same-window growth of the main rivals

Surge (WIFI) customers, millions



Surge in one view

+1.98M homes in 12 months: 3.0× (1.2M reading) to 7.8× (0.8M reading) IndiHome's underlying growth.

Take-up ~61% of homes passed (fiber), ~57% (FWA); ~165k adds a month.

2026 guidance **2.1M fiber + 1M FWA** (cut from 3M), ~8M by 2028; blended ARPU **Rp70–108k**.

Moving up-market: **Prime 500 Mbps Rp168k**, **Super 2 Gbps Rp250k**, Combo fiber + FWA.

Who competes for the home connection

Rival	Model	Price (monthly)	Measured experience	Scale
Surge: Starlite, Internet Rakyat	Fiber + licensed 1.4 GHz FWA	Rp100k for 200 Mbps fiber or 100 Mbps FWA, VAT incl.	Not in Opensignal panel	2.36M (Jun 2026)
Indosat: HiFi, HiFi Air	Fiber + self-install FWA	HiFi 100 Mbps Rp345k*; HiFi Air 125 GB Rp100k	Best consistent quality (69.7%)	~400k homes; 1M HiFi Air target
XLSmart: XL SATU, First Media	Fiber + mobile (converged)	Joy 100 Mbps Rp350k*; XL Home Biz Rp419k	Best reliability (496.8)	~1.0M fixed
Biznet	Fiber specialist	100 Mbps Rp375k; ~Rp150k in small cities	Fastest download (41.0 Mbps)	City by city
Oxygen.id / MyRepublic	Fiber; MyRepublic Air FWA	Nova 100 Mbps Rp482k*; Air from Rp100k	Fastest upload (29.6 Mbps)	Merging
IndiHome (Telkomsel)	Fiber, widest reach	100 Mbps Rp270k (Jakarta Selatan, app)	Reliability 351.1, 7th of 8	9.513M
Informal and satellite	RT/RW Net; Starlink	Very cheap / premium remote	n/a	Tens of thousands of resellers

** older press prices; VAT basis unclear. The low-price front is fiber more than wireless; the quality front is regional (XL SATU in West Java, Kalimantan, Sulawesi; HiFi in Jakarta; GlobalXtreme in Bali-Nusra).*

Source: Surge disclosures (S48, S49, S58, S60, S70); Indosat (S29, S68); XLSmart (S16, S69); Opensignal (S14); price comparisons (S28); Telkomsel app (S45)

Appendix 8

Measured experience: Opensignal award slots and IndiHome's urban profile against seven rivals

Award slots won by IndiHome



Rural wins outside Maluku–Papua: all five in Banten and Yogyakarta, some in five more regions; Banten rural has IndiHome as the only ISP measured.

Rank and gap to the leader (8 ISPs)

National reliability (points)	**351.1**	496.8 (XL SATU)	**−29%**
National consistent quality	**57.8%**	69.7% (Indosat HiFi)	**−11.9 pts**
National download	**26.2 Mbps**	41.0 (Biznet)	**−36%**
National upload	**12.9 Mbps**	29.6 (Oxygen.id)	**−56%**
National video (0–100)	**66.0**	70.3 (Oxygen.id)	**−4.3 pts**
Urban consistent quality	**58.8%**	69.9%	**−15.9%**
Urban download	**26.4 Mbps**	40.8	**−35.3%**
Urban upload	**13.3 Mbps**	30.5	**−56.4%**

Reading: averages reflect the plan mix (many low-speed legacy plans), but reliability and consistency depend less on plan speed. The gap splits between the network (group fiber) and the home (router, Wi-Fi coverage, ONT), where Telkomsel can act alone and already sells fixes (ONT upgrades, Ultra Mesh, FTTR).

Source: Opensignal, Indonesia Fixed Broadband Experience, 1 Jan–31 Mar 2026 (S14); award slots counted by script

Appendix 9

Digital mystery shop: what five Telkomsel product pages tell a household about which product fits

IndiHome	Orbit	EZnet	EZnet Wireless	Telkomsel One
Cross-product guidance None: "Cek Jangkauan" returns IndiHome plans only **Who it says it is for** Device counts per card (1–6 … 1–12) **What it shows** Double Speed 200 on 100 Mbps Rp280k … 500 on 300 Mbps Rp500k; **180 or 360 days**; Rp50k installation promo **Notable** The boost is time-limited; plan names mix boosted and base speeds	**Cross-product guidance** None: "lihat modem Telkomsel Orbit lainnya" points to other Orbit modems **Who it says it is for** "Paket modem WiFi spesial **untuk semua**" **What it shows** Devices Rp445k–Rp2,339k; 30-day packs 45 GB Rp82k to 300 GB Rp410k; prices vary by location **Notable** No target household named	**Cross-product guidance** One path: FAQ says upgrade by moving to IndiHome **Who it says it is for** "Internet rumah murah dan mudah **untuk semua**" **What it shows** Card 20 Mbps Rp170k; FAQ still 10 Mbps Rp150k; ex-VAT; 12 months, Rp1M exit fee **Notable** Contradicts itself; shows an expired promotion	**Cross-product guidance** None: no comparison with Orbit, IndiHome or EZnet **Who it says it is for** "Fleksibel tanpa kontrak jangka panjang" **What it shows** 50 Mbps Rp170k + admin fee, ex-VAT; 350 GB then 5 Mbps; modem lent; postpaid **Notable** The opposite model to Orbit for the same household job	**Cross-product guidance** None: one line for the whole brand **Who it says it is for** "Untuk keluarga" **What it shows** Dynamic Rp270k / Rp350k; Complete Rp405–730k; Easy Rp120k; installation Rp150k; VAT basis not stated **Notable** No Double Speed; device for the Orbit quota not stated
✖ no routing tool	✖ no routing tool	✖ no routing tool	✖ no routing tool	✖ no routing tool

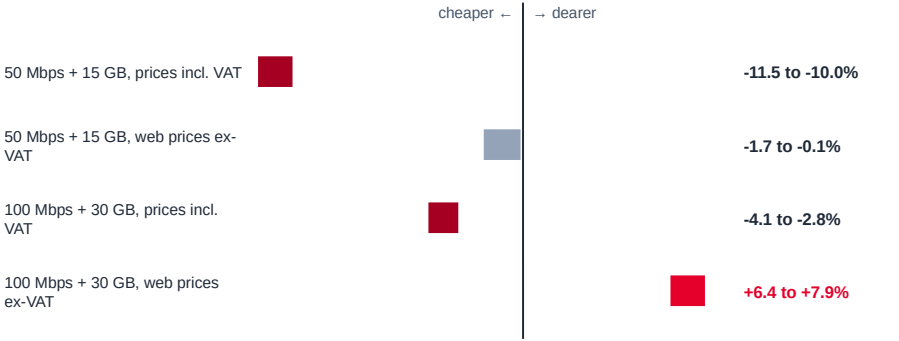
So what: none of the five pages routes a household between products; two say "for everyone"; two wireless products sell the same job on opposite models; prices appear on four VAT bases; bundle installation terms change by campaign (Rp150k vs a Rp50k standalone promo).

Source: Telkomsel web pages read 10 Oct 2026: /indihome, /orbit, /eznet, /eznet-wireless, /one (S65); MyTelkomsel app (S45)

Appendix 10

Spend: the price ladder, Telkomsel One against its parts, and what Double Speed lends

Telkomsel One against IndiHome + a mobile pack, by VAT reading (% cheaper or dearer)



Robust in both readings: at best a small saving, for less speed than standalone IndiHome with Double Speed, and a Rp150k installation fee against Rp50k standalone.

From charging for extras to giving them away

Period	What Telkomsel did with extras	Effect
2023–2024	Triple-play as the norm; ARPU Rp253k → Rp238k	Bundled extras carried ARPU
2025	Internet-only spreads; "Higher Speed More Price"; tiered OTT	ARPU −10.1%; mix moves to 1P
2026	"High Speed Same Price": Double Speed for 180 or 360 days; Viu, Prime Video, Vision+ included	Speed and some content free; Netflix +Rp30k, FTTR +Rp160–295k still paid

What the base plan now contains

180 / 360 days — Double Speed on 100–150 Mbps / 200–300 Mbps bases: a renewal cliff at month 6 or 12 unless designed as a choice

≈Rp109–132k — retail value of the three services in a Double Speed plan, 45–55% of the Rp240k price est.

−2.5% to +8.4% — what the average home pays against the cheapest standalone plan, by base

Rp27.0k — one speed step ex-VAT (Rp240k → Rp270k incl. VAT), the unit the boost-end choice prices

So what: the things that could raise household spend (convergence, whole-home Wi-Fi, safety, content, a second location) are priced as add-ons per product, lent for a period, or not sold as household outcomes. Management chose HSSP "to defend premium base"; the issue is that its end is not designed.

Source: Telkomsel app and web (S45, S46, S65); user-reported Super Seru pack prices (S47); OTT retail prices (S67, weak); Telkom 1H26 Info Memo (S04); our calculation

Appendix 11

Usage crossover: monthly cost of each option by usage, and how robust the Orbit result is

What each option costs a household a month (Rp k); cheapest option shaded

Monthly use	Orbit	HiFi Air	EZnet Wireless	EZnet fiber (card; FAQ)	IndiHome entry	Internet Rakyat	Starlite fiber 200 Mbps	Mobile data only
25 GB	66	75	188.7	188.7 (166.5)	240	100	100	78
50 GB	100	75	188.7	188.7 (166.5)	240	100	100	155
100 GB	176	100	188.7	188.7 (166.5)	240	100	100	310
150 GB	232	150	188.7	188.7 (166.5)	240	100	100	465
250 GB	352	200	188.7	188.7 (166.5)	240	100	100	775
450 GB	642	250	188.7, slowed to 5 Mbps above 350 GB	188.7 (166.5)	240	100	100	1,395

IndiHome incl. VAT (app); EZnet and EZnet Wireless Rp170k ex-VAT ×1.11 (EZnet Wireless adds admin and installation fees); Internet Rakyat and Starlite incl. VAT; Orbit and HiFi Air VAT basis not stated; mobile data at Rp3.11 per MB.

Does the Orbit result depend on the weak 450–500 GB figure? A conservative family build-up

Component	Basis	GB a month
Phones on home Wi-Fi	4–6 members × 12.5 GB per Telkomsel customer	50–75 est.
TV streaming	2 hours a day × 30 days × 1–3 GB an hour (Netflix SD–HD)	60–180 est.
Family home, conservative		**≈110–255** est.

Reading
· Within Telkomsel, Orbit is cheapest only up to **95 GB** (vs EZnet's FAQ price) or **125 GB** (vs Rp188.7k).
· Above ~155 GB Orbit costs more than IndiHome; at 450 GB **2.7×**.
· EZnet Wireless already sells heavy wireless use at Rp0.54k per GB.
· So Orbit fits light, flexible or temporary homes: the issue is positioning, and the technology works. An unlimited Orbit plan at Rp150k (2025 press) was not seen on 10 Oct; Phase 0 checks it.

Source: Telkomsel app and web (S45, S65); HiFi Air (S29); Starlite and Internet Rakyat (S60, S70); data yield (S04); Netflix (S59); PMK 131/2024 (S66); our calculation

Appendix 12

Households: needs, product fit and what each household type is worth to Telkomsel

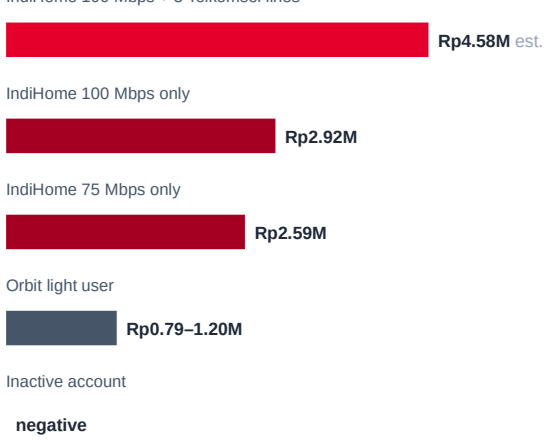
Product fit by household need: ✔ good · ~ partial · ✖ poor

Need (size)	Orbit	EZnet Wireless	EZnet fiber	IndiHome	Telkomsel One	Best-fit rival today
First connection, tight budget (most of ≈58–60M)	~	~	~	✖	~	Starlite / Internet Rakyat
Flexible or temporary living (9.29M non-owners)	✔	~	✖	✖	~	HiFi Air
Connected families (4–5 people: 46.5%)	✖	~	~	✔	~	HiFi / XL SATU; Starlite Prime
Performance: gaming, hybrid work (~68M gamers)	✖	✖	✖	~	✖	Biznet / Oxygen.id
Large or premium homes (≈12.9M)	✖	✖	✖	✔	~	XL SATU; Starlite Super
Fiber-dark or remote homes	~	~	✖	✔	~	Starlink
Home micro-business (66M MSMEs)	✖	✖	✖	~	✖	XL Home Biz

Every need has a Telkomsel product that fits at least partly, but the best fit differs by need, two wireless products compete for the same homes, and nothing routes a household to the right one.

Pain points along the journey: five brands with no "which one is for me" · coverage check shows IndiHome only · four VAT bases · installation Rp150k bundle vs Rp50k standalone, Orbit device Rp385k+ · below rivals on measured experience in cities · a speed cliff when a boost ends · Rp100k unlimited options within reach.

Revenue to Telkomsel a year (Rp M, ex-VAT)



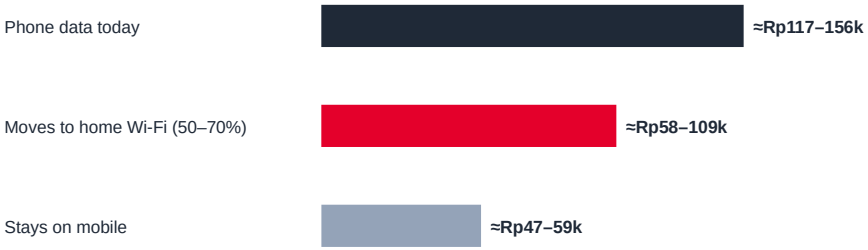
Jobs households describe
"Keep everyone online at once." · "Connect us now, without a contract, while we rent." · "Stop topping up phone data." · "Keep the kids safe online." · "Let me work or game without lag." · "Make the Wi-Fi reach every room."
A converged family is worth **≈1.6–1.8×** a fiber-only home; an inactive account destroys value.

Source: File 02 §8: BPS via GoodStats (S50, S51); tSurvey in Telkomsel AR 2025 (S03); Telkom 1H26 Info Memo (S04); IESPA and Kadin (S54); our calculation and estimate

Appendix 13

Offload: what a first home connection is worth to Telkomsel, and one household's path in rupiah

A 3–4-line Telkomsel family, Rp k a month (est.)



Who connects the home	Effect on Telkomsel per family a month
A rival (e.g. Rp100k Starlite)	**Loses ≈Rp58–109k** of mobile data revenue, gains nothing
Telkomsel at a Rp100k price (Rp90.1k ex-VAT)	**Net ≈−Rp19k to +Rp32k**
Telkomsel at IndiHome's entry price (Rp216.2k ex-VAT)	**Net ≈+Rp107k to +Rp158k**, if the family pays it

Offload happens whoever connects the home, so winning the home protects the home revenue and, by Telkomsel's account, the family's lines; it does not protect phone-data spend.

One household's path, contribution a month (est.)

Step	Revenue vs mobile-only	Contribution vs mobile-only	Contribution vs a rival connecting
Orbit start (Rp90.1k)	+Rp6.5k (Rp90.1k − Rp83.6k offload)	**−Rp24.9k**	**+Rp19.4k**
Moves to IndiHome (Rp216.2k)	+Rp132.6k	**+Rp2.2k**	**+Rp46.5k**
Adds one line on the household plan		**+Rp19.8k**	**+Rp19.8k**

Orbit's own economics: per 100k Orbit starts, **≈37.9k** move to fiber within five years at the base conversion rate, each paying back its trade-in and installation through the higher fiber price. An Orbit start breaks even against a mobile-only family at a **49%** margin (the model uses 21.5%).

So what for the solution

· The home check targets Telkomsel families **a rival is about to connect**: they pay from month one.
· Connecting a family that would stay mobile-only pays only once it reaches IndiHome and adds lines.
· The value comes after the first connection: ladder steps, family lines and homes kept.

Source: Telkom 1H26 Info Memo, data yield and payload (S04); file 02 §5.5; file 03 §8.7; model output; our estimate

Appendix 14

Root cause: issue tree, five whys and fishbone converge on one governing cause

Five whys: three routes to one mechanism

Through pricing: why did value per household fall?	Through targets: why did inactive accounts pile up?	Through channels: why are rivals winning the new homes?
ARPU −15.6% in two years; per paying home −3.5% in two quarters after the stock stopped growing	0.8–1.2M accounts brought in no revenue and were removed	Rp100k fiber and FWA undercut on price; better-rated fiber wins in cities
Why? Mix moved to internet-only and entry tiers; speed and content added at the same price	**Why?** Acquired on activation-paid incentives, postpaid with no deposit, mostly outside the mobile relationship	**Why?** IndiHome's margin is thin and urban experience trails; Orbit priced for light use
Why? Each product's price point defended against Rp100k by giving more (HSSP)	**Why?** Growth targets set in lines (800k–1M a year), with no paying-home test	**Why?** Roles follow network and tier; no offer built around a household leaving phone data
Why? Each product judged on its own ARPU and line count	**Why?** Inherited line economics: fiber bought per line, growth reported in lines	**Why?** The mobile relationship was not the acquisition channel
Why? No one measured on what the household pays across products and over time	**Why? No household-level measure for targets and incentives to follow**	**Why? Fixed and mobile run and counted separately; no one owns the household with both**

Governing cause: Telkomsel runs households on inherited line economics (targets in lines, KPIs per product, fiber bought per line) while the market has repriced a line at Rp100k. No one is measured on what a household is worth across products and over time.

Fishbone	Product	Price	Place and channel	People and process	Technology	Measurement	Policy and market
Lands in	Role and entry	Spend; role	Entry; role	Entry; delivery	Delivery and cost	Governing cause	Context

Source: File 02 §11: issue tree on the household value equation, five whys, fishbone

Appendix 15

Gap analysis and governing cause: current against ideal, tested against Telkom's "Maximize LTV"

#	Dimension	Current condition	Ideal condition
1	Product roles	Roles by tier and network; no routing tool on 5 pages; Orbit and EZnet Wireless sell one job two ways; Orbit counted as mobile	Every home routed by need and location, with a visible path between products
2	Entry terms	0.8–1.2M inactive accounts (59–88% of 18 months' net adds); activation-paid incentives; postpaid, no deposit	Every new account a paying household, on terms that surface non-payment early
3	Entry channel	1H25 net growth almost all in homes without a Telkomsel line; 26–36M unconnected homes are Telkomsel families est.	New homes come mainly through the mobile relationship
4	Value per household	Average home −2.5% to +8.4% vs the entry price; speed lent 180–360 days	Spend rises through things the family values and pays for
5	Convergence	59% → 65% overlap, mostly a denominator effect; bundle at best a small saving	Converging is the better deal; depth is measured
6	Paths	EZnet → IndiHome only in an FAQ; boost expiry not designed	Each household has a designed next step
7	Experience	1 of 50 urban award slots; 7th download, 8th upload, 4th consistent quality	At or above rivals where they compete
8	Cost to serve	Per-line wholesale fee on a ~40%-utilised network; Orbit heavy packs at average cost per GB est.	Technology and price matched to usage and spare capacity

Is it an org chart or a measurement problem?

Evidence it is measurement: Orbit counted as mobile, IndiHome as fixed, EZnet inside IndiHome · targets in lines (800k–1M a year) · convergence measured as overlap · fiber leased per line.

Evidence against an org-chart problem: one VP for home broadband and FMC marketing already exists.

Falsification test: would a household-managed Telkomsel show…

Expected sign	In the 2026 plan	On the web
Routing in its channels	✖	✖
A bundle that rewards converging	✖	✖
Targets in paying homes	✔ billing-linked incentives	✖
Score	**1 of 3**	**0 of 3**

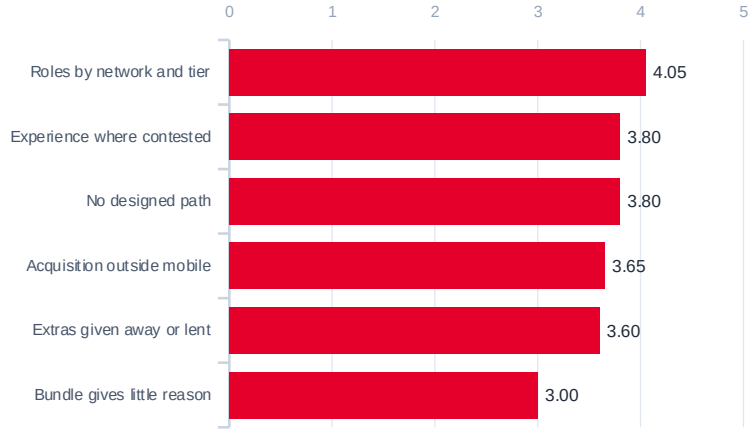
Telkom already names **"Maximize LTV"** for IndiHome; the levers and published KPIs still work product by product. If stores and 188 route consistently (Phase 0 mystery shop), the cause narrows to measurement.

Source: File 02 §11.1, §6.7; Telkom 1H26 Info Memo p.43, p.47 (S04); Telkomsel web (S65); Bisnis on targets (S62)

Appendix 16

Prioritisation: one value tree, scoring rules and how the ranking holds under 69 tests

Weighted score by driver (0–5)



Unscored (size unknown): entry terms 3.15–4.35; cost not matched to usage 2.20–3.40.

Score band	5	4	3	2
Significance (year-3 value)	≥ Rp1T	Rp0.25–1T	Rp0.1–0.25T	< Rp0.1T
Urgency	Exploited now	≤ 12 months	1–3 years	None
Controllability	Own levers	Mostly own	Shared	Outside
Evidence	Primary, quantified	Primary, partial	Estimated	Hypothesis

Driver values and scores (weights 40 / 25 / 20 / 15)

Driver	Year-3 value est.	Sig.	Urg.	Ctrl.	Evid.	Score
Roles by network and tier	Rp64–235bn	3	5	5	4	**4.05**
Experience where contested	Rp464–696bn	4	4	3	4	**3.80**
No designed path	Rp264bn	4	3	5	3	**3.80**
Acquisition outside mobile	−Rp22bn to +Rp256bn	3	4	5	3	**3.65**
Extras given away or lent	Rp154bn	3	4	4	4	**3.60**
Bundle gives little reason	Rp92bn	2	3	5	3	**3.00**

Does the ranking survive? 69 tests

Test	Top driver	Role and entry branch first?
5 weight sets	Roles in 4 of 5	5 of 5
48 one-point changes	Roles alone in 41	48 of 48
12 two-band significance changes	Roles in 7 of 12	12 of 12
4 cases of acquisition value	Roles in all 4	Yes
Unscored drivers at their extremes (20)	n/a	17 of 20 (3 exceptions, significance-heavy)

Source: File 02 §13: year-3 run-rate of a 5% step on each driver's own lever; scores by stated rules; sensitivity by script

Appendix 17

SWOT with a so-what and the TOWS moves mapped to H, O and M

Strengths

Widest reach; only scale ISP in Maluku–Papua (30 of 30) · 153.5M mobile customers; 26–36M of their families unconnected est. · largest new 2.6 GHz block (80 MHz) · one bill and MyTelkomsel · CVM data at scale · group fiber ~40% utilised + ~1.2M released ports · EZnet Wireless proves cheap heavy wireless

Weaknesses

No cross-product routing; nine overlapping offers · acquisition outside the mobile base; 0.8–1.2M inactive · 1 of 50 urban award slots, upload last · Orbit quota-priced, device upfront, "for everyone" · bundle saves little · thin margin, wholesale dependence est. · four VAT bases · capital rationed

Opportunities

≈58–60M unconnected homes; ≈36M aspiring middle est. · policy 50% by 2029 · household outcomes (Wi-Fi, safety, gaming) · idle fiber to refill · migration paths (Globe benchmark) · 3.345M homes without a Telkomsel line · new spectrum from 2027 · the boost expiry as a designed moment

Threats

Surge: 2.36M fixed customers, ~165k adds a month · HiFi Air and MyRepublic Air at ~Rp100k · quality-led fiber rivals in cities · regulatory price pressure · weaker 2026 consumer confidence · speed is a commodity · cheap fiber moving up-market · a rival connecting a Telkomsel family takes ≈Rp58–109k a month est.

So what: Telkomsel's rare strengths are household assets (reach, the family's phones, spectrum, data, idle fiber); its weaknesses sit in how homes are routed, acquired, priced and rewarded; the threats are price-led. Respond fit-led and relationship-led, with what rivals need years to copy; a price move is the one response they can always undercut.

TOWS: the option space, mapped to the pillars

SO	Use the mobile base and data to reach families already spending on phone data	H
SO	Reach, idle fiber and new spectrum for fiber-dark and first-time homes, with a path to fiber	H
SO	Sell household outcomes on top of reach	O
ST	Make convergence the reward that keeps the family	O
ST	Where rivals are absent, invest in quality and hold price	M
ST	Defend mobile families before a rival connects their home	H
WO	One routing logic across five products; distinct roles for Orbit and EZnet Wireless	H
WO	Turn the boost expiry into a priced upgrade moment	O
WO	Refill idle fiber through migration	M
WT	Avoid a Rp100k headline price war; acquire only paying homes	H
WT	Close the measured gap where rivals are strongest, inside the home first	M
WT	One VAT-inclusive price everywhere	H

Source: File 02 §15; Opensignal (S14); Telkom 1H26 Info Memo (S04); Surge (S48, S58, S60); MPA (S24); Komdigi (S21)

Appendix 18

Product roles and the routing tree: the job of each product and the five questions in order

Product	Its job	The household	The promise	How it differs (not price)
Orbit	**The flexible start**	New, moving, renting or light homes (up to ~125 GB a month)	Connected today; no installation; take it with you	Portable; no contract; credit toward fiber when the home settles
EZnet Wireless	**The wireless home for heavy use**	Heavy-use homes with no fiber, or not staying 12 months	A large allowance (350 GB) on a monthly plan, modem lent	Fixed location and a large allowance, unlike Orbit's portable packs
EZnet	**The settled light fiber home**	Homes staying 12+ months, with fiber and light use	Unlimited fiber at entry speed	The first step on the fiber path
IndiHome	**The family's home hub**	Settled homes with heavy use, TV streaming, work or gaming	The whole home connected: every room, safe for children, fixed fast	Whole-home Wi-Fi, parental controls, fix-time promise, performance tiers
Household plan (Telkomsel One, reframed)	**What ties the roles to the family**	Families with 2+ Telkomsel lines on any home product	One bill, shared family quota, household services	Each added line adds quota and services; standalone installation terms

The check, in a fixed order

Step	Question	Answer → next
1	Stay at this address 12+ months?	No → wireless only · Yes → step 2
2	Fiber available, with a spare port?	No → wireless · Yes → fiber
3	How much will the home use?	From consented phone data, or a short quiz
4	How many devices; how big is the home?	6+ devices → Wi-Fi check, mesh or FTTR
5	How many Telkomsel lines?	2+ → household plan

How a household moves between them

- **Mobile-only family, rival nearby** → Orbit start → EZnet or IndiHome when use > ~125 GB or it settles
- **Renting, heavy use** → EZnet Wireless → IndiHome when it settles and fiber arrives
- **Settled, fiber, light use** → EZnet → IndiHome as use and family grow
- **Any home + 2 Telkomsel lines** → Household plan: shared quota, safety, one bill
- **Any home about to leave** → A better-fitting plan, offered only to keep it

Source: File 03 §3–4; usage crossover (Appendix 11); Telkomsel web (S65); case book (S01)

Appendix 19

Initiatives and owners: what each pillar changes, who delivers it and when it starts

H · Household fit	O · Offer a path up	M · Make staying better	E · Evaluate household value
Household-fit check in every channel **Build Nov 2026–Jan 2027; pilot 1H27**	Designed paths: Orbit → fiber with modem trade-in, EZnet → IndiHome, single → family **2H27**	Home Wi-Fi promise: Wi-Fi check, mesh for 75k at-risk homes a year, fix-time promise **Pilot 1H27; contested cities 2028**	One household identity (consent) **1H27**
Stated roles; fewer overlapping offers **Phase 0 quick win**	Boost-end choice: keep the speed for a fee, take an outcome pack, or return to base, with 30 days' notice **Randomised test from Jan 2027**	Retention triggers with a 10–20% random hold-out **Pilot 1H27; national 2H27**	Household KPIs and targets; an Orbit baseline **2027 KPI contracts**
Targeted, opt-in home check for Telkomsel families **Pilot 1H27**	Household plan: each added line adds shared quota and services **2H27**	Port fill and capacity-aware routing with the group **2028 (not in the model's value)**	One owner of household value; shared bonus pool **Named in Phase 0**
Pay sales on homes paying at 3 and 12 months **1H27**			

Who does what (RACI, short form)

Initiative	Accountable	Delivers	Consulted
Fit check; stated roles	VP Home Broadband & FMC Consumer Marketing	Digital channels; product marketing	InfraNexia (port data); legal
Home check; triggers; paths	VP Household Consumer Business Strategy Management	CVM	Data protection officer
Pay on paying homes	Head of sales and channels	Sales operations	Finance
Boost-end choice; household plan	VP Home Broadband & FMC Consumer Marketing	Billing and IT; product marketing (FMC)	Legal (consumer notice); mobile marketing
Wi-Fi promise; port fill	Network and CPE operations	Field operations; InfraNexia	Finance
Household identity, KPIs, owner	Household P&L owner (CEO office)	IT and data; finance	Telkom reporting

Source: File 03 §5, §7.5; VP titles as reported (S11, S30, S62); other functions generic

Appendix 20

Gates and KPIs: pass marks at 60% of plan, randomised tests and targets for 2028 and 2031

Gate ladder: one unit for model and gate

Gate	KPI (model's unit)	Plan	Pass mark	Measured by
Gate 1 · Jun 2027	Expiring homes choosing to pay at boost end	8%	**≥ 4.8%**	National randomised test
	Home-check opt-in × conversion	10% × 10%	**≥ 0.6%**	Pilot areas
	New homes paying at 3 months	quality	**≥ 95%**	Pilot billing
	Save rate on triggered homes vs hold-out	10%	**≥ 6%**	Hold-out
	EZnet → IndiHome / single → family steps a year	3% / 0.5%	**≥ 1.8% / 0.3%**	Randomised offer vs hold-out
	Lines added a year in homes without one	2%	**≥ 1.2%**	Randomised offer vs hold-out
Gate 1b · Dec 2027 / Jun 2028	Churn in Wi-Fi-promise and triggered homes vs hold-out	−1pp	**≥ −0.6pp**	50k–150k homes per arm
Gate 2 · Dec 2028	Household KPIs monthly; Orbit baseline; programme EBITDA run-rate	Rp65bn (2029)	**≥ 0**	Finance; cost cap in tranches

Why 60%: if every take-up input lands at **55% of plan**, the life-of-homes value is zero; at 60% it is **+Rp45bn**. The pass mark is joint (switching off any one lever still leaves the value positive). At the pass mark the 5-year NPV is −Rp197bn; it reaches zero only at **112% of plan**. If Gate 1 fails, national spend stops with about **Rp98bn** of cash spent.

KPI targets (net stocks after churn) est.

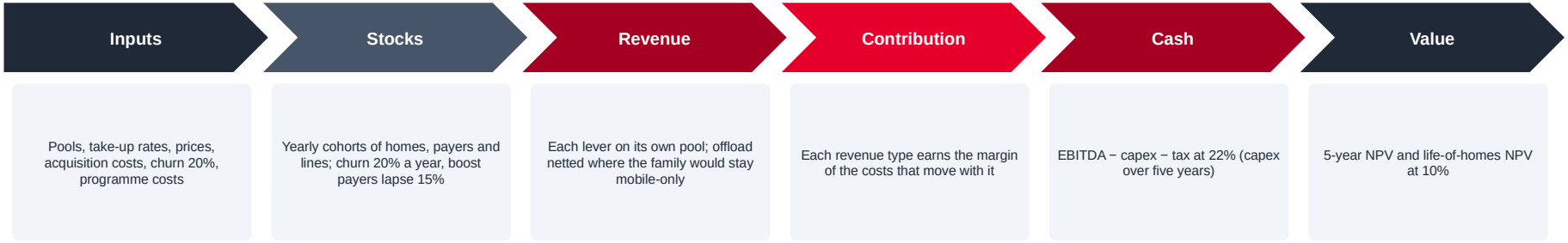
KPI	Today	2028	2031
Paying homes won through fit	0	73k	**213k**
Homes kept	0	59k	**204k**
Homes paying for a chosen extra at boost end	0	114k	**114k**
Telkomsel lines added in homes without one	0	67k	**197k**
Orbit starts moved to fiber	0	1k	**22k**
EZnet → IndiHome and single → family steps	0	90k	**267k**
Household value per paying home	Not measured	Baseline	Rising

The case's own impact terms (est., 2031)
· Subscriber conversion: **22k** Orbit homes on fiber; **267k** ladder steps
· ARPU uplift: **Rp415bn** a year from paths, boost-end choice and the plan
· Retention: **204k** homes kept; **−1pp** churn in contested cities
· FMC adoption: **197k** lines added
· Incremental revenue: **Rp1,134bn** a year

Source: File 03 §7.3–7.4; model-v4 output: 55% of plan = zero life value, 60% = +Rp45bn

Appendix 21 (1/2)

Financial model: structure, comparator and the contribution logic



Contribution margins by revenue type

Revenue type	Margin	Why
New fiber homes and homes kept	**21.5%**	Household EBITDA margin; the fiber fee is paid per line
Orbit starts	**21.5%** (15–49% tested)	Conservative placeholder; no per-line fiber fee
Upgrades on an existing line	**70%**	Little extra cost; network ~40% utilised
Content, TV and outcome packs	**40%**	Net of content revenue share
Added mobile lines	**53%** less 10% for plan benefits	Mobile EBITDA margin

Comparator, horizon and ramps

Do-nothing comparator: the household base stays at today's run-rate, boosts revert silently, no price change; continued erosion would favour the plan.

Horizon: programme years 2027–2031 carry new homes, take-up and all costs; run-off 2032–2041 keeps only homes already won as they churn off (no new homes, IT running cost only): a conservative alternative to a perpetual terminal value.

Ramps follow the plan: fit 26.5% in 2027, 80% in 2028, 100% from 2029; paths, plan and triggers from 2H27; Wi-Fi promise 5% in 2027, 60% in 2028; the boost-end test covers 75% of the 2027 legacy cohort.

Money: Rp bn ex-VAT; discount rate 10% (9–12% tested); 2027 tax credit offset against Telkomsel's profit.

Source: File 03 §8.1; model-v4 (base case identical to model-v3)

Appendix 21 (2/2)

Financial model: inputs, the pool each applies to, test values and anchors

Input	Pool it applies to	Base	Test values	Basis
Home-check invitations a year	Telkomsel families without fixed broadband (26–36M)	6.0M	3.0–9.0M	assumption: a fifth of the pool a year
Opt-in × conversion	Invited families	10% × 10%	5–15% each	assumption; measured in the pilot
Better fit with other families	Non-Telkomsel new homes (1.28M × 47.5%)	+2 points	1–3	assumption; market growth (MPA)
Home-check families who would stay mobile-only	Home-check homes	30%	20–50%	assumption; targeted at rival areas
Start product and price	Fit homes	50% Orbit Rp90.1k; 50% fiber Rp190k	Orbit Rp59.5–120k; fiber Rp170–216.2k	Telkomsel prices, ex-VAT
Acquisition cost: Orbit / fiber via mobile / other channels	New homes	Rp150k / 400k / 750k	100–250 / 250–600 / 500–1,000	assumption; no public figure
Offload per affected family	Mobile-only counterfactual homes	Rp83.6k a month	Rp58.3–108.9k	estimate
Routing-down leakage	500k normal new homes a year	2%	1–10%	assumption; price fences
Boost cohorts expiring	Legacy 2027 / recurring from 2028	20% of base / 500k a year	10–35% / 250–750k	assumption; sized in Phase 0
Boost-end paid take-up / lapse	Expiring homes / payers	8% / 15% a year	4–12% / 10–25%	assumption; randomised test
Orbit → fiber a year; trade-in	Orbit starts	15%; Rp150k + Rp50k refurbishment	8–25%	assumption
EZnet → IndiHome / single → family a year	EZnet homes (15% of base) / the base	3% / 0.5%	1.5–5% / 0.25–1%	assumption; steps Rp46.2k / Rp125k
Lines added a year	3.345M homes without a Telkomsel line	2%	1–3%	assumption
Wi-Fi promise churn effect	Contested cities, 35% of base	−1pp	−0.5 to −1.5pp	assumption; Gate 1b
Triggers: reach × would-leave × save	The base	10% × 40% × 10%	reach and save 5–15%	assumption; hold-out
Annual churn; discount rate	All stocks; cash flows	20%; 10%	15–25%; 9–12%	assumption
Capex 2027–2031	Fit check Rp80bn, data platform Rp80bn, mesh Rp135bn	**Rp295bn**	data Rp60–120bn; mesh Rp500–800k a home	bottom-up estimate
Running costs	Marketing Rp40bn (half reallocated), IT Rp15bn, office Rp5bn a year	≈Rp60bn a year	marketing Rp25–75bn	bottom-up estimate

Source: File 03 §8.2–8.3; Telkom 1H26 Info Memo (S04); Telkomsel app and web (S45, S65); MPA (S24); BRI Danareksa (S61)

Appendix 22

Financial results: yearly cash flow, scenarios and the gate valued as an option

Base case by year (Rp bn) est.

Year	Fit homes (net)	Homes kept (net)	EBITDA: fit	Paths	Boost-end	Household plan	Keep	Programme costs	EBITDA	Capex	Free cash flow	Cumulative
2027	19k	11k	−4	3	10	2	−1	−86	**−75**	80	**−135**	−135
2028	73k	59k	−10	20	20	10	8	−60	**−11**	95	**−96**	−231
2029	131k	119k	−4	52	20	22	35	−60	**65**	55	**6**	−225
2030	177k	166k	5	84	20	34	62	−60	**145**	55	**71**	−154
2031	213k	204k	12	111	20	43	84	−60	**210**	10	**167**	**12**

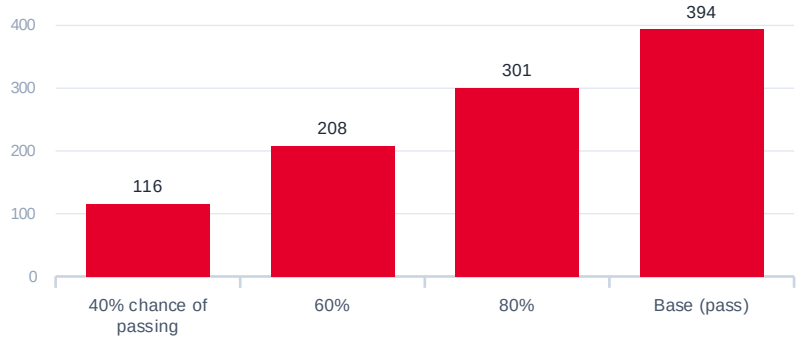
2031 revenue **Rp1,134bn**: fit Rp328bn less Rp49bn of offloaded mobile data · paths Rp278bn · boost-end Rp37bn · household plan Rp100bn · keep Rp441bn, about 4.4% of today's ≈Rp25.6T household revenue. Tax: −20 (2027), −10, 4, 19, 33. Components are rounded in the model output, so some rows differ from their totals by Rp1bn.

Scenarios (Rp bn)

Measure	Base	Every input at its worse end (ungated)	Every input at its better end
5-year NPV (2027–2031)	**−46**	−618	1,301
NPV over the life of the homes won	**394**	−655	3,597
Cash payback	**2031**	Not reached	2028
Lowest cumulative cash	**−231** (2028)	−823	n/a
IRR / B/C, life-of-homes cash flows	**35.1% / 2.95**		

If Gate 1 fails: Rp98bn of cash (PV −Rp69bn after tax); at most Rp133bn if all 2027 capex and training had gone first.

The gate as an option: expected life-of-homes NPV

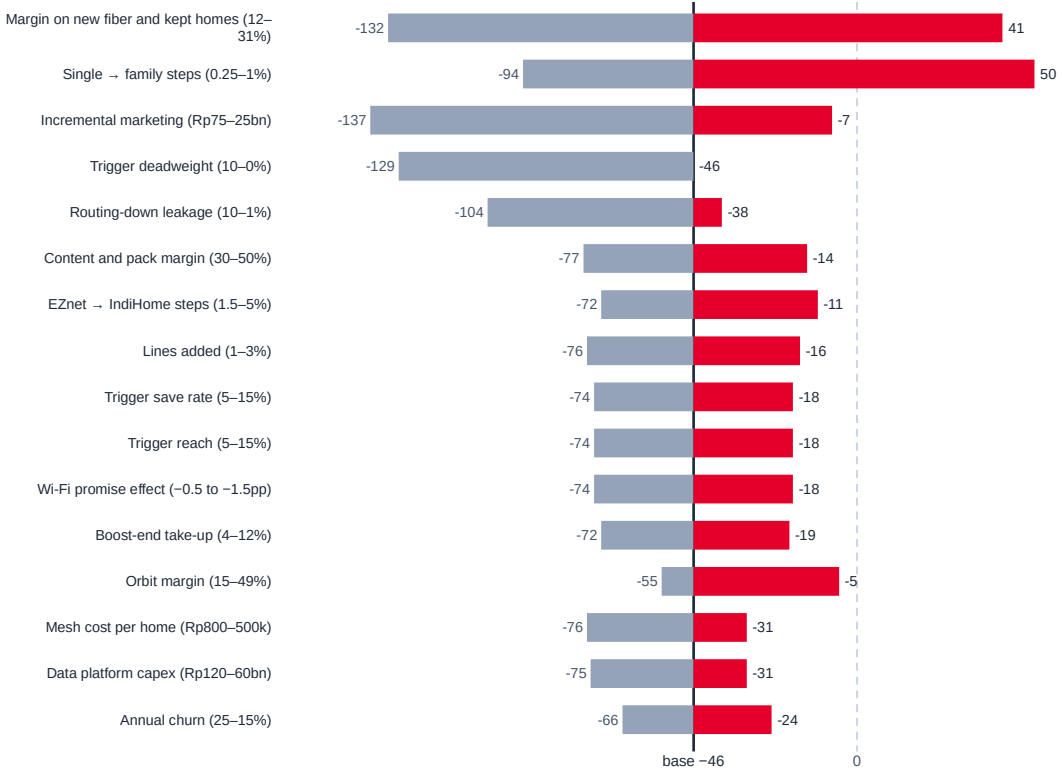


Source: Model-v4 output (file 03 §8.4–8.5); every figure is an estimate on stated assumptions

Appendix 23

Sensitivity: the inputs that move NPV most, the break-evens and the known biases

Tornado on the 5-year NPV (Rp bn), sorted by absolute swing



Break-evens (5-year NPV = 0, one input at a time)

Input	Plan	Break-even
Boost-end take-up	8%	14.9%
Trigger save rate	10%	18.2%
Wi-Fi promise effect	1pp	1.82pp
Margin on new and kept homes	21.5%	25.8%
Lines added a year	2%	3.54%
EZnet steps a year	3%	5.62%
Single → family a year	0.5%	0.74%
Incremental marketing	Rp40bn	Rp22bn

Life-of-homes view and known biases

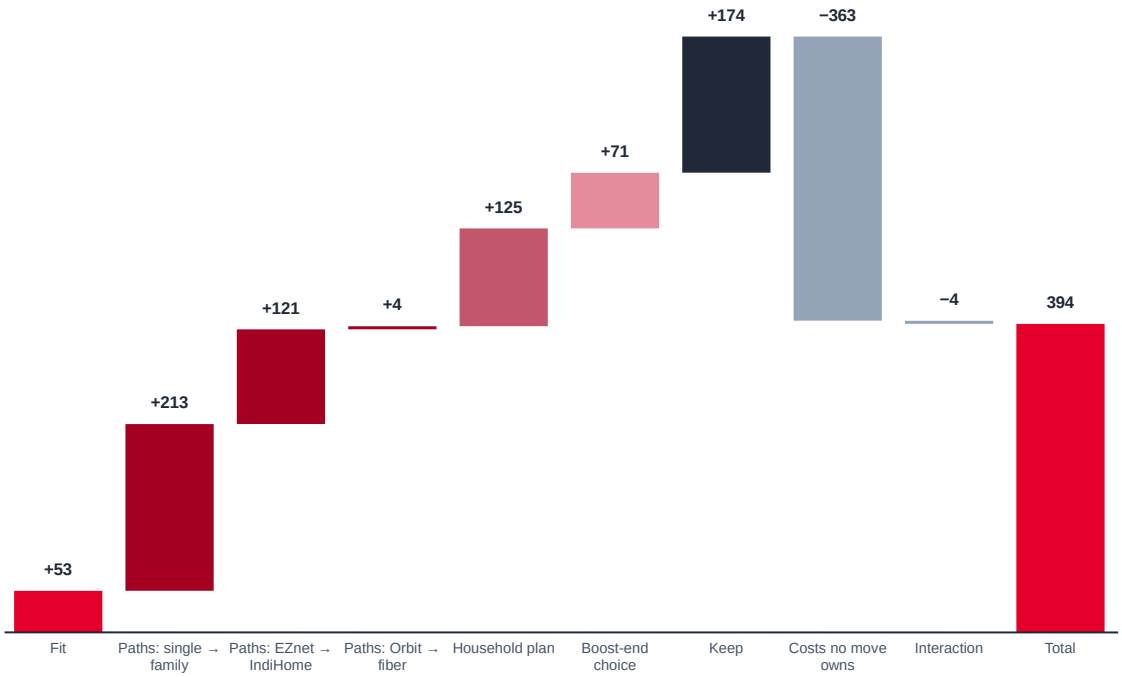
· Value stays positive down to a **4.2%** margin and to **55% of plan** on every take-up at once.
· Conservative: Orbit at the fixed-line margin (at 35%, life NPV **Rp435bn**).
· Optimistic: 5% deadweight and saved homes churning at 30% → **Rp272bn**; half of step-ups would happen anyway → **Rp227bn** (Rp164bn with half the lines too). Step rates are therefore measured as lift over a hold-out.
· Leakage breaks the case only at **28%**.

Source: Model-v4 output: tornado on the 5-year NPV (base −Rp46bn), each input alone at its worse and better test value; break-evens one input at a time

Appendix 24

Value split: what each pillar is worth, the reconciliation, and the price-led comparator

From the moves to the total (life-of-homes NPV, Rp bn) est.



Reconciliation

Item	Rp bn
Fit	53
Paths (ladder 334 + Orbit 4)	338
Household plan	125
Boost-end choice	71
Keep	174
Sum of moves	**761**
Costs no move owns (platforms, marketing, office)	−363
Interaction	−4
Total	**394**

The price-led alternative, in one line
Every **Rp10k** off IndiHome across the base costs **≈Rp1,142bn a year**, nearly all of it margin because a price cut saves no cost: over five times the **Rp210bn** of EBITDA a year this plan adds by 2031. Holding price while meeting the Rp100k front with Orbit's flexible start is the core of "without relying primarily on price competition".

Source: Model-v4 output: value = life-of-homes NPV lost with the move switched off; reconciliation; price comparator (9.513M × Rp10k × 12, ex-VAT)

Appendix 25 (1/2)

Risks: twelve risks with early warnings, triggers and the actions they set off

#	Risk	Likelihood · impact	Early warning	Trigger → action	NPV at stake
1	Routing down cannibalises revenue	M · H	Revenue per paying home, pilot vs control	Leakage > 5% → tighten price fences; Orbit not shown to settled heavy homes with fiber	Rp46bn life value at 5%
2	Boost-end backlash	M · M	Complaints, churn in the test arm	Take-up < 4.8% at month 3 → outcome packs as default; never a silent cut	Rp53bn
3	Rival price response	H · M	Churn where a rival builds	Rival build detected → triggers and Wi-Fi checks there first; no headline cut	Rp56bn
4	Data protection (PDP law)	M · H	Opt-in rate; complaints	Opt-in < 5% → simplify consent; no location data without a lawful basis	Home-check volume
5	Wholesale cost makes budget fiber uneconomic	M · M	Incremental cost per budget home	Above ≈Rp90k → budget homes to Orbit only	Not modelled
6	Teams resist household KPIs	H · H	KPI adoption in reviews	Not in 2027 KPI contracts → Phase 0 does not exit; escalate	All levers
7	Billing or IT slips	M · M	Test start date	Slip > 1 month → run the test by SMS on one cohort	Rp53bn
8	InfraNexia port data withheld	M · M	Agreement signed in Phase 0	Not signed → availability only; EZnet where field teams confirm ports	Fit volume
9	Agents ignore the fit check	M · M	Share of agent sales through the check	< 60% → incentives only on sales made through it	Fit volume
10	Offload larger than assumed	H · M	Mobile data revenue per converged family	> Rp100k a month → target rival areas only	Rp48bn
11	Capital rationing	M · L	Gate decisions	Capex in tranches; mesh can slip a year	Rp45bn (mesh)
12	Reallocated marketing not released	M · M	Budget transfers signed in Phase 0	Not signed → P&L owner funds within the cap; roll-out waits	Rp130bn swing

Source: File 03 §9; NPV at stake from the model-v4 tornado

Appendix 25 (2/2)

Feasibility and trade-offs: the constraint check and how the six objectives are balanced

Constraint check

Constraint	Verdict	Why
Capital	Feasible	Rp295bn capex over five years, after the gates; 2025 free cash flow Rp22.5T, financing outflows Rp22.9T, so spend is phased
Regulation	With care	Consent-based offers (PDP law); one VAT-inclusive price; notice before a boost ends
Wholesale and group	Partly ours	Port-fill terms need the group; the plan works without them
Technology	Feasible	Channel logic; household linking on existing CVM analytics
Organisation	Hardest part	One P&L owner, KPI contracts, shared bonus pool
Time	Feasible	Quick wins in weeks; boost-end test from Jan 2027; national by 2028
Competition	Defensible	Fit, family relationship and reach are harder to copy than a price; XL SATU can copy convergence

Trade-offs between the case's six objectives, and how the plan sets them

Acquisition vs margin	Home Wi-Fi offloads phone data, so the home check targets families a rival is about to connect (they pay from month one)
Affordability vs monetisation	Orbit gives a no-commitment start; price fences keep routing-down leakage under 5% of new homes
Retention vs monetisation	Cheaper plans only to save a leaving home; a random hold-out measures the deadweight
Experience vs capital	Mesh only for 75k at-risk homes a year in contested cities, released in tranches after Gate 1b
Differentiation vs simplicity	Five products keep five distinct jobs; the nine overlapping offers are cut back
Growth vs prudence	Gates at 60% of plan: a failed pilot stops at about Rp98bn of cash

Source: File 03 §10, §2–§9; Telkomsel AR 2025 cash flow (S03)

Appendix 26

Theme: how the solution creates lasting value responsibly, with a target for each commitment

Fair terms

Sales paid on homes that keep paying; no silent end to a boost

KPI: Share of new homes paying at 12 months

Target: ≥ 90% by 2028

Price transparency

One VAT-inclusive price everywhere

KPI: Channels quoting the same price (mystery shop)

Target: 100% by the end of Phase 0

Digital inclusion

A first connection without commitment; wireless homes where fiber is absent

KPI: Orbit starts and the first-time share

Target: ≈36k Orbit starts a year est.; first-time share in the fiber-dark pilot ≥ 60% of the Phase 0 estimate

Child online safety

Parental controls in the household plan (86% of 1,000 surveyed interested)

KPI: Homes with active parental controls

Target: 25% of household-plan homes by 2029

Circularity

Orbit modems traded in and refurbished on the path to fiber, beside 83.49 t of ONTs refurbished in 2025

KPI: Devices refurbished

Target: Every traded-in modem (≈22k converted homes in 2031 est.)

Data protection

Consent-based household identity and offers

KPI: Opt-in rate; data complaints

Target: Reported quarterly; zero data findings

Lasting value

Household lifetime value as the operating metric

KPI: Household value per paying home

Target: Baseline 2027; rising every year

Do well and do good point the same way. Homes on the right product, on fair terms, stay longer and step up: in the model, keeping homes and the paths up are worth **Rp512bn** over the life of the homes.

Source: File 03 §6.2; file 02 §19; tSurvey in Telkomsel AR 2025 p.12 (S03); modem refurbishment p.159 (S03); PDP law UU 27/2022

Appendix 27 (1/2)

Assumptions register: every assumption, its value, rationale and how Phase 0 will measure it (1/2)

#	Assumption	Value (test range)	Rationale	Measured by
1	Super Seru promo packs represent what a family pays for separate mobile data	15 GB ≈Rp60–65k; 30 GB ≈Rp90–95k	Match the Rp3.1k per GB yield	Another pack benchmark
2	Telkomsel's mobile EBITDA margin	50–56%	Splits household from mobile margin (12–31%)	Segment disclosure
3	When the inactive stock built up	Three timing scenarios	Bounds the paying-household restatement	Cohort data
4	Family size and Telkomsel lines	4–6 members; 3–4 lines	Usage build-up, offload, household value	Phase 0 survey
5	TV streaming time in a family home	2 hours a day, SD to HD	Usage build-up	Survey; Orbit usage
6	Content in Double Speed plans worth its retail price	Prime Video Rp59k, Viu Rp30–53k, Vision+ Rp20k	Content inside the plan	Included tiers; willingness to pay
7	People's class shares apply to households	8.6 / 24.4 / 49.3 / 17.3 / 0.5%	Class sizing	Household-level class data
8	EZnet fiber is unlimited at 20 Mbps	No quota	Usage crossover	EZnet terms
9	Scoring rules and weights	40 / 25 / 20 / 15	Prioritisation (tested 69 ways)	Sensitivity, Appendix 16
10	Share of a family's phone data that moves to home Wi-Fi	50–70%	Offload	Phase 0 survey
11	Annual churn of the base	15–25% (20% base)	No public benchmark	Churn disclosure
12	Telkomsel families' share of homes	45–60%	45% = share of mobile subscriptions	Survey; household data
13	Home-check invitations a year	6.0M (3–9M)	About a fifth of the 26–36M pool	Phase 0 app reach
14	Opt-in and conversion of invited families	10% and 10% (5–15%)	No benchmark; conservative for a free check	Pilot
15	Better fit raises the win rate with other families	+2 points (1–3)	Small effect of the check alone	Pilot vs control
16	Home-check homes that would stay mobile-only	30% (20–50%)	Home check targets rival areas	Pilot targeting data

Source: File 02 §21.1; file 03 §13. These are the only numbers in the deck that are not sourced or calculated from sources

Appendix 27 (2/2)

Assumptions register: every assumption, its value, rationale and how Phase 0 will measure it (2/2)

#	Assumption	Value (test range)	Rationale	Measured by
17	Acquisition cost per home by route	Rp150k / 400k / 750k	No public figure; the household buys the Orbit device	Pilot cost per paying home
18	Routing-down leakage	2% (1–5%) of 500k new homes a year	The tree routes down only light users	Pilot revenue per home
19	Boost cohorts expiring	20% of base in 2027; 500k a year from 2028	HSSP began in 2026 with 180/360-day boosts	Phase 0 cohort data
20	Paid take-up at boost end; lapse; fee-induced churn	8% (4–12%); 15%; 0 (0–1%)	A clear choice with notice	Randomised test
21	Orbit → fiber a year; trade-in; refurbishment	15% (8–25%); Rp150k; Rp50k	Usage-triggered offers with a modem credit	Pilot
22	EZnet share of base; EZnet → IndiHome; single → family	15%; 3% (1.5–5%); 0.5% (0.25–1%)	Small, usage-triggered steps	Pilot
23	Lines added in homes without a Telkomsel line	2% a year (1–3%)	Plan benefits, no discount	Pilot
24	The do-nothing comparator	Base at run-rate; silent reverts; no price change	Static; continued erosion would favour the plan	n/a
25	Contested cities' share of base; churn effect	35% (25–45%); −1pp (−0.5 to −1.5)	Urban areas where IndiHome trails	Gate 1b hold-out
26	Triggers: reach, would-leave, save rate; offer cost	10%, 40%, 10%; Rp150k a save	Standard retention logic	Hold-out
27	Margins on upgrades and content	70% (60–80%); 40% (30–50%)	Spare capacity; content share	Finance
28	Discount rate (WACC)	10% (9–12%)	Not published for Telkomsel	Telkom disclosure
29	Programme costs	Bottom-up (Appendix 21)	Channel logic, data linking, targeted mesh	Vendor quotes; budgets
30	Probability of passing Gate 1	40–80%	Shown as a range	Gate 1
31	Theme targets	Appendix 26	Set as commitments	Annual review

Source: File 02 §21.1; file 03 §13. These are the only numbers in the deck that are not sourced or calculated from sources

Appendix 28

Considered and parked: five ideas kept for later, and why each waits

Orbit as the household's "satellite"	**Why promising:** A child's kos room or campus, a worker away from home, the mudik trip, a second home, on the household plan with IndiHome as the hub	**Why parked:** No data on multi-location families	**Next step:** Test with Phase 1 families; KPI: Orbit devices per IndiHome home
Win-back of removed homes on fair terms	**Why promising:** 0.8–1.2M homes already wired; Telkom names "targeted retention & reactivation"	**Why parked:** They stopped paying once; terms must change first	**Next step:** Phase 2 pilot after pay-on-paying-homes
Paid whole-home add-ons	**Why promising:** FTTR, mesh rental, safety packs: ≈Rp183–337bn a year per 1% FTTR attach	**Why parked:** Device cost and take-up unknown	**Next step:** Price test in performance and premium homes
Government housing programme	**Why promising:** 3 million homes: volume and inclusion	**Why parked:** Price-capped; terms unknown	**Next step:** Watch
Lower churn on the family's mobile lines	**Why promising:** Telkomsel reports "significantly lower churn propensity" for converged homes	**Why parked:** No number published	**Next step:** Measure through household KPIs

None of these is counted in the model; each would add value.

Source: File 03 §14, §8.9; Telkom 1H26 Info Memo p.36 (S04); file 02 §4.4

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Method note: figures tagged in the team's working files as primary, secondary, calculated, estimate or assumption; estimates are marked "est." on every page; assumptions are listed in Appendix 27.